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Impact of Pressure Pulsations on Diesel Air Path Control and Compensation Measures

Master of Science Thesis

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A Master Thesis Report on

Impact of Pressure Pulsations on Diesel Air Path Control

by
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Performed in the **PTE Diesel Department** at



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Abstract

Stricter pollution norms have led to increased efforts to achieve lower emissions. Exhaust Gas Re-circulation (EGR) is a well known measure to reduce NO_x internally in the engine. In order to achieve better performance of an engine in terms of emissions it is very essential to control the flow of exhaust gases through EGR valve. Generally, an Engine Control Unit (ECU) uses mean value models or map-based approaches to calculate the mass flow through EGR valve.

The main driving factor of the mass flow through an EGR valve i.e. the exhaust gas pressure is continuously fluctuating throughout an engine cycle. This continuous change in the pressure lead to variations in the mass flow through EGR valve, thus introducing inaccuracies in the desired flow through EGR valve, thereby affecting the engine performance greatly. Hence it is very essential to consider the pulsations in a combustion cycle.

In this thesis an EGR valve model is developed in Matlab/Simulink environment. The model is able to consider the effect of exhaust pressure pulsations and is found to be more accurate than the mean value models used in the ECU.

Acknowledgements

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Stockholm, March 2019

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1 Introduction

The environmental effects of the pollutants from vehicles opened a new field of sustainable development and antipollution campaigns. Today in Europe all the diesel vehicles must meet the regulations of Euro 6 emission standard, which poses stringent limits on the emissions, especially on particulate matter and the NOx. Figure no.1 below shows the limits posed by Euro 6 norm on the emission of the PM and NOx.

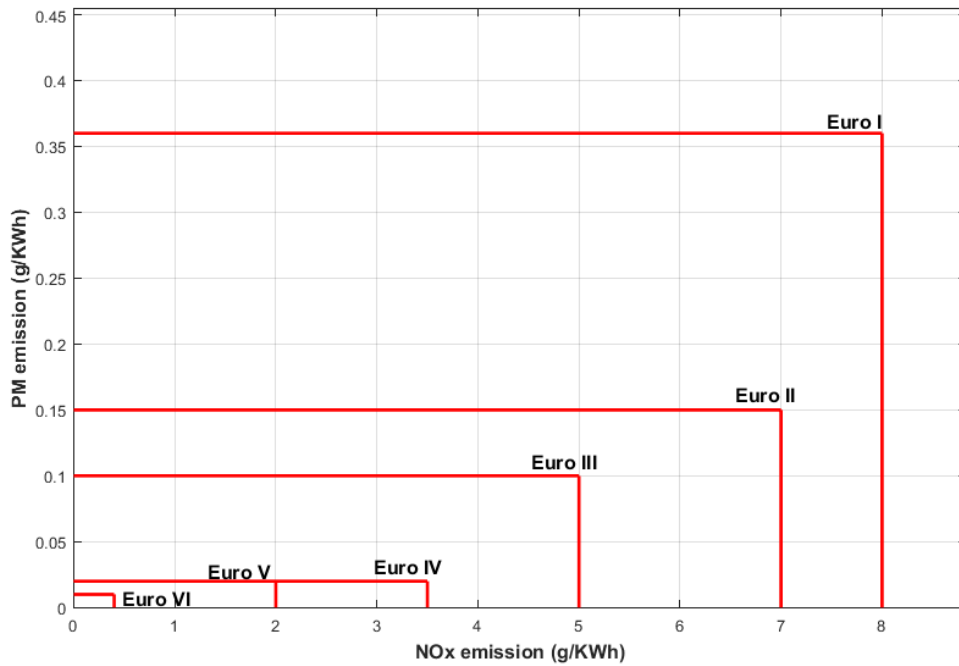


Figure 1: European emission norms

NOx is generally produced by the reaction of nitrogen and oxygen during the combustion of fuels. NOx forms in the combustion zones with high oxygen concentration and high combustion temperature. One of the ways to reduce the NOx emissions is to use ‘NOx storage catalyst’ in the after treatment system. It is also possible to reduce the generation of NOx during the combustion process by lowering the flame temperature. Exhaust Gas Recovery (EGR) is a technique which is used to lower the combustion temperature. A portion of exhaust gases is recirculated into the intake manifold. The exhaust gas acts as an inert gas during the combustion process and absorbs some amount of the heat of combustion, thus making the combustion process slower and thereby reducing the combustion temperature leading to the reduction of NOx. Though EGR helps in reducing NOx emissions it can lead to increase in other pollutants like hydrocarbons (HC), carbon monoxide (CO) and carbon dioxide (CO₂). Increase in the EGR amount also leads to increased fuel consumption [3]. Hence it is very important to choose accurate amount of EGR mass going towards an

engine, to achieve good engine performance in terms of emissions and fuel consumption. The flow of EGR is controlled by an air path control system as shown in figure 2.

Some amount of exhaust gas flow after coming out from an engine is directed towards the intake manifold through a throttle valve which is also called as an EGR valve. The EGR mass flow going towards the IM can be controlled through this valve. The restriction of EGR valve is administered by an ECU.

The ECU uses mean value models to determine the mass flow through the EGR valve. But since the flow on the engine exhaust side is pulsating, the mass flow determined by ECU is different from the mass flow which is actually going towards IM. This difference in the calculated and actual mass flow, can lead to increased emissions and fuel consumption. Hence it is very essential to consider the influence of pulsations on the mass flow through EGR valve.

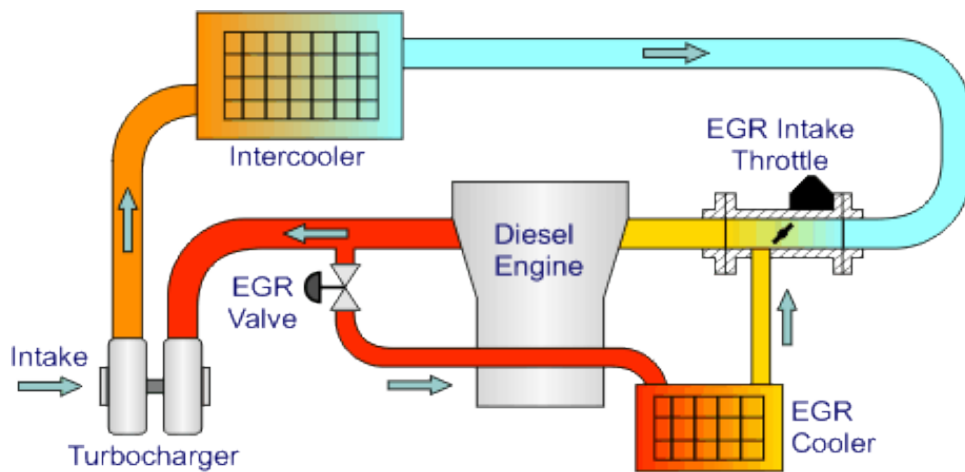


Figure 2: EGR Path [2]

The purpose of this master thesis is to analyse the effect of exhaust pressure pulsations on the mass flow through an EGR valve. Also to develop an EGR model which is able to calculate the mass flow with the consideration of pressure pulsations and is real time capable enabling its use for ECU integration.

1.1 Problem formulation

As mentioned in the previous section, the flow on an engine exhaust side is highly pulsating. A typical variation of exhaust pressure is as shown in figure 3. The figure shows the variation of exhaust pressure for one engine cycle, with the values of pressure recorded each crank angle degree rotation of engine.

The Electronic Control Unit (ECU) of an engine calculates the mass flow through the EGR valve by considering the inputs of upstream and downstream pressures, and the upstream temperature across the EGR valve.

Since the ECU needs to be operated real time, it cannot consider the cycle to cycle variations occurring in the pressures and temperatures of exhaust flow. Hence, modern ECU's use mean-value models to achieve high speed of operation.

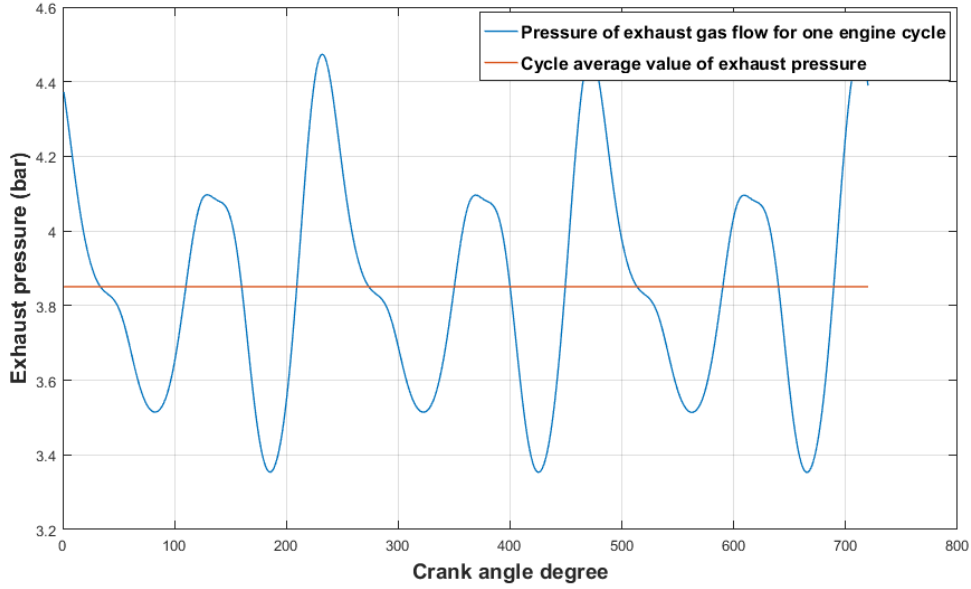


Figure 3: Pressure variation on engine exhaust side

The mean-value models, as the name suggests, consider the ‘mean’ or average values of the upstream and downstream pressures and temperatures across EGR valve for the calculation of mass flow through it. Thus, it can be easily derived that the exhaust mass flow that is actually going towards the intake manifold varies from the one calculated by ECU. This error in the calculation of mass flow generated due to the use of mean value models can affect the engine performance greatly as mentioned in the previous section.

In some engine layouts, the pressure on the engine intake side becomes more than the engine exhaust side, due to turbocharger operation. In short, the pressure on the downstream side of EGR valve becomes more than the upstream side, which might lead to back-flow through the EGR valve. In order to stop this back-flow, a reed valve (or non return valve) is used along with EGR valve. The reed valve blocks the exhaust gases from flowing in the negative direction, through the EGR valve. The use of reed valve has some drawbacks, which include increase in the amplitude of pressure fluctuations across EGR valve. This increase in the amplitude further adds to the error in the mass flow calculation caused by mean value models.

In order to eliminate this error in the calculation of mass flow, it is very essential to develop an EGR model which has following characteristics:

1. The model should incorporate the cycle to cycle variations of pressures and

temperatures in mass flow calculation.

2. The EGR mass flow calculated by the model must be accurate.
3. It must be fast enough to be able to work real time.

This thesis deals with development of an EGR model for ECU, which has above mentioned abilities. General layout of this report is mentioned in next section.

1.2 Report layout

1. Reference for EGR valve model

This section gives information about the reference used in this thesis for the development of EGR model.

2. EGR valve model

This section gives all the information about the procedure followed while developing an EGR valve model. It mentions the equations used in the development of the model, along with detailed explanation of all the individual components used in the model. The influence of pressure ratio on the mass flow through the EGR model is also analysed in this section.

3. Influence of constant inputs on the mass flow through EGR valve

The effect of considering the mean (cycle averaged) values of the inputs of EGR model instead of crank angle resolved (i.e. values considered each crank angle degree of crankshaft) values is analysed on the mass flow.

4. Modelling of upstream pressure as a sinusoidal signal

The effect of modelling the upstream pressure input of EGR model as a sinusoidal wave while considering the mean values for all other inputs, on EGR mass flow is analysed. The analysis is further extended with the consideration of harmonics in the sinusoidal upstream pressure. The effect of amplitude of upstream pressure on the mass flow is also analysed.

5. Development of real time solution

A real time capable model of EGR valve is developed, which is able to consider the effect of pulsations and is able to calculate the mass flow accurately.

2 Reference for EGR valve model

An AVL CruiseM model of a six cylinder engine is used as a reference in this work. It is intended to develop an EGR model in MATLAB/Simulink environment, which is able to calculate the same EGR mass flow as that of CruiseM engine model. The Simulink model thus developed will further undergo several modifications, thereby becoming a real time capable EGR model.

Following procedure is followed in order to set up a reference:

1. The AVL CruiseM model of a 13 liter, six cylinder heavy duty diesel engine is simulated at the operating points shown in figure 4 below. During the simulation at each operating point, the values of mass flow, upstream- downstream pressures and temperatures, actual and stoichiometric air-fuel ratios across the EGR valve of the engine are recorded for each degree of crankshaft rotation. The values are recorded for one engine cycle, i.e. for 720° . Thus, the recorded data is crank angle resolved. The CruiseM engine model and the location of its EGR valve is shown in figure 8.

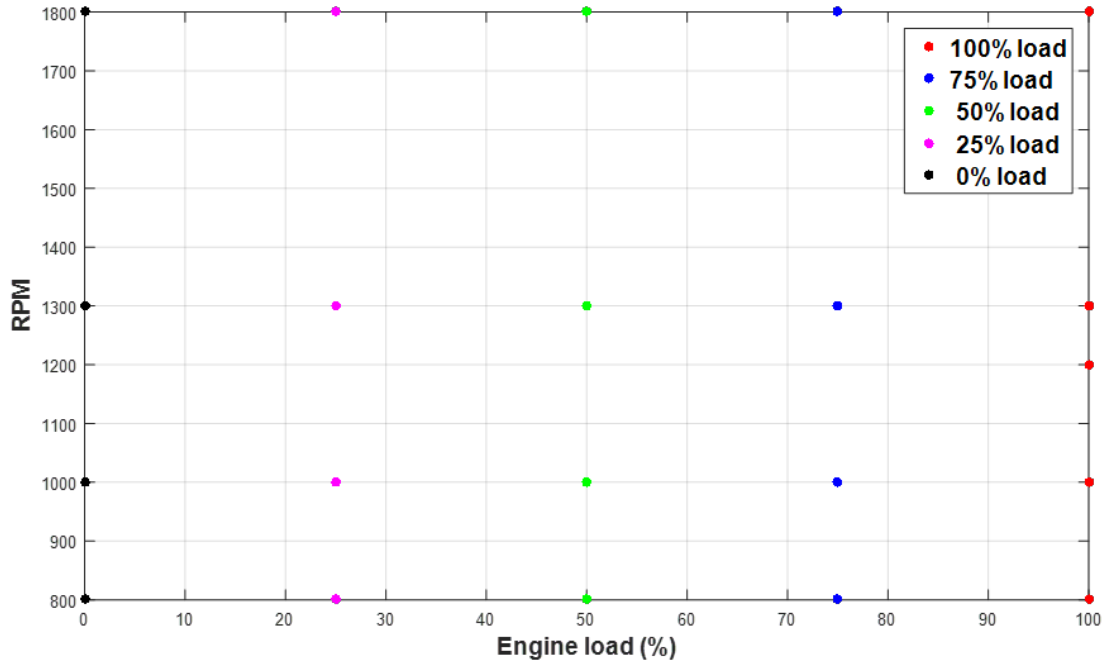


Figure 4: Operating points selected for the simulation

2. The CruiseM engine model uses Transient Momentum Balance (TMB) [4] [5] to calculate the mass flow through the EGR orifice. However, TMB cannot be used for calculation of mass flow in a real time model [6]. Hence, in order to set up a reference which is suitable for a real time models, it is very essential to determine the mass flow through EGR orifice with the use of throttle equation as shown by equation 1.

3. Thus, to obtain an EGR mass flow with the throttle equation, a separate AVL CruiseM orifice model representing the EGR valve of the 6-cylinder engine is constructed as shown in figure 5 below. The Restriction 1 block in the figure represents the EGR valve and the blocks System Boundary 1 and 2 represent the upstream and downstream conditions of the EGR valve respectively. In further sections this model will be addressed as a CruiseM EGR model.



Figure 5: CruiseM EGR model representing the EGR valve of the 6-cylinder engine

4. The CruiseM EGR model is then fed with the upstream and downstream values of pressures and temperatures, which are recorded from the CruiseM engine model at a particular operating point as mentioned in the bullet point 1 above. These inputs of pressures and temperatures are given to the system boundary 1 and 2 blocks as shown in figure 5.
5. The Transient Momentum Balance option from the properties of Restriction 1 block in figure 5 is unchecked as shown in figure 6 below. An important point to be noted here is when the TMB option is unchecked, the CruiseM EGR model uses the throttle equation as given by 1 to calculate the mass flow through the EGR valve.

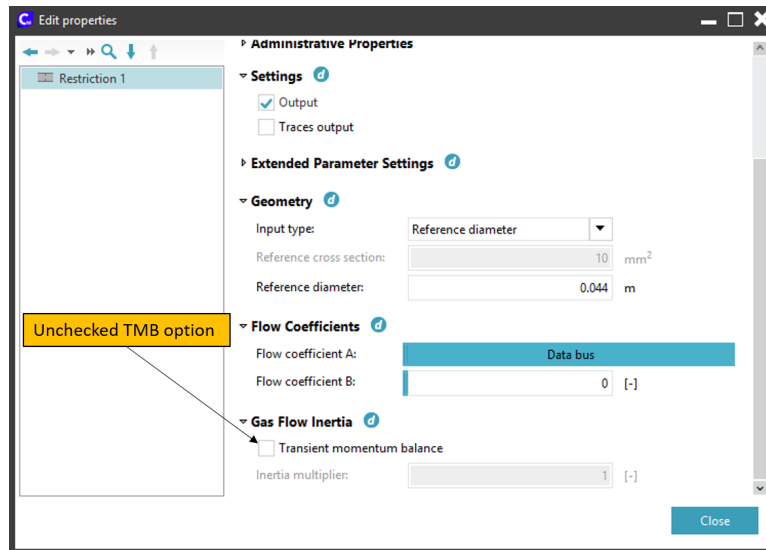


Figure 6: Dialog box of the properties of Restriction block in CruiseM

6. With the boundary conditions set and the TMB option unchecked, the CruiseM EGR model is simulated. The values of EGR mass flow are recorded from the

model. The data of mass flow thus obtained from the EGR model is CA resolved. This process is repeated for all the operating points selected in figure 4.

7. Thus, the data of EGR mass flow is recorded from the CruiseM EGR model, while the data of upstream-downstream pressures and temperatures, and air-fuel ratios is recorded from CruiseM engine model shown in figure 8. All this data of mass flow, pressures, temperatures and air-fuel ratios corresponding to a particular operating point is stored in a excel sheet.

A typical waveform of upstream and downstream pressure and temperature recorded from CruiseM engine model at the operating point of 100% load and 1300 RPM every crank angle degree are as shown in figure 7 below.

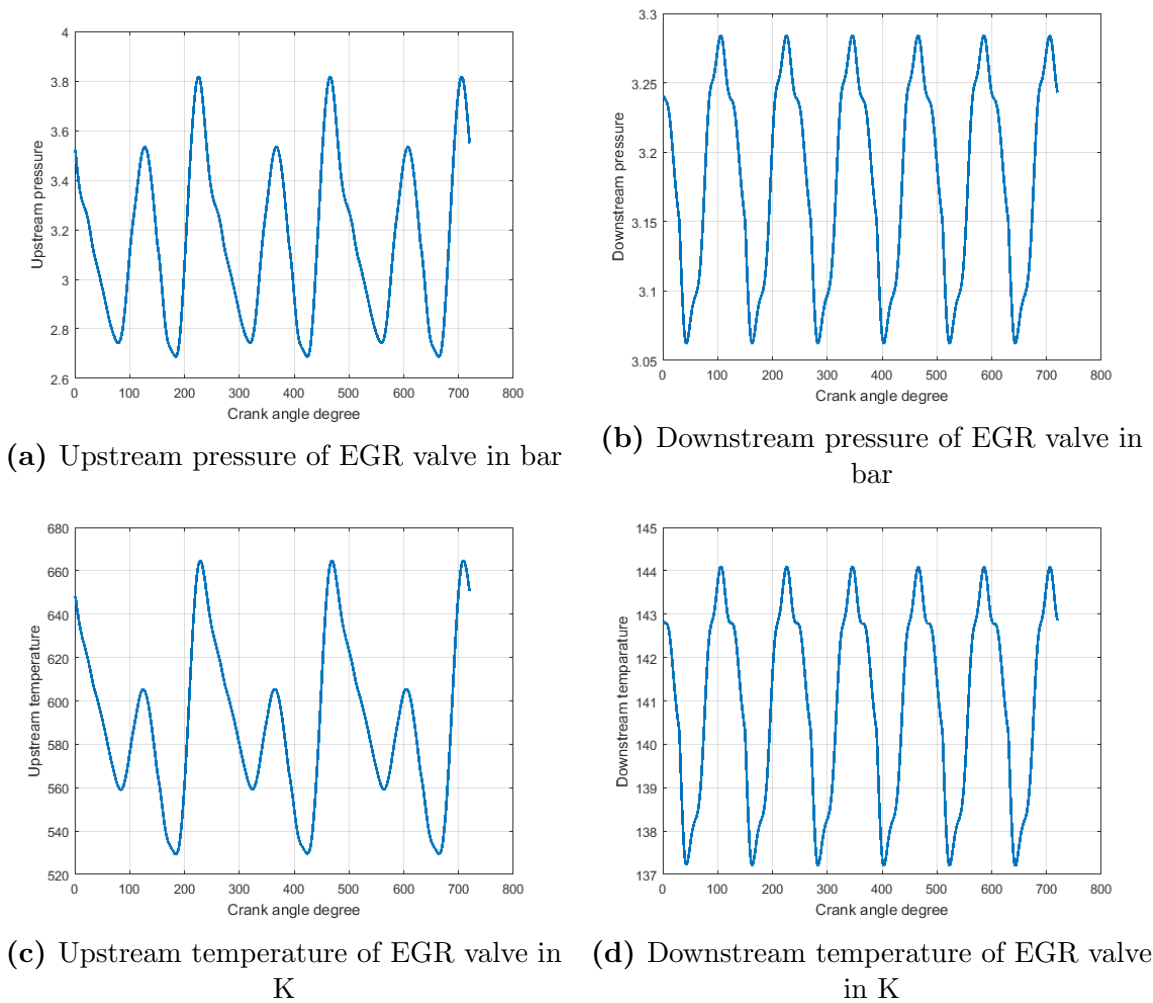


Figure 7: Pressures and temperatures recorded from CruiseM

8. A detailed model of the flow across EGR valve is constructed in Matlab/Simulink environment. This model is then simulated with the data of pressure, temperatures, air-fuel ratios, recorded from CruiseM EGR and engine models as mentioned

above. Detailed procedure of development of Simulink EGR model is explained in section 3.

9. The mass flows of simulink and CruiseM EGR models are then compared. For the simulink model to be accurate, its mass flow must be the same or should have less than 1% difference from the mass flow of CruiseM EGR model.

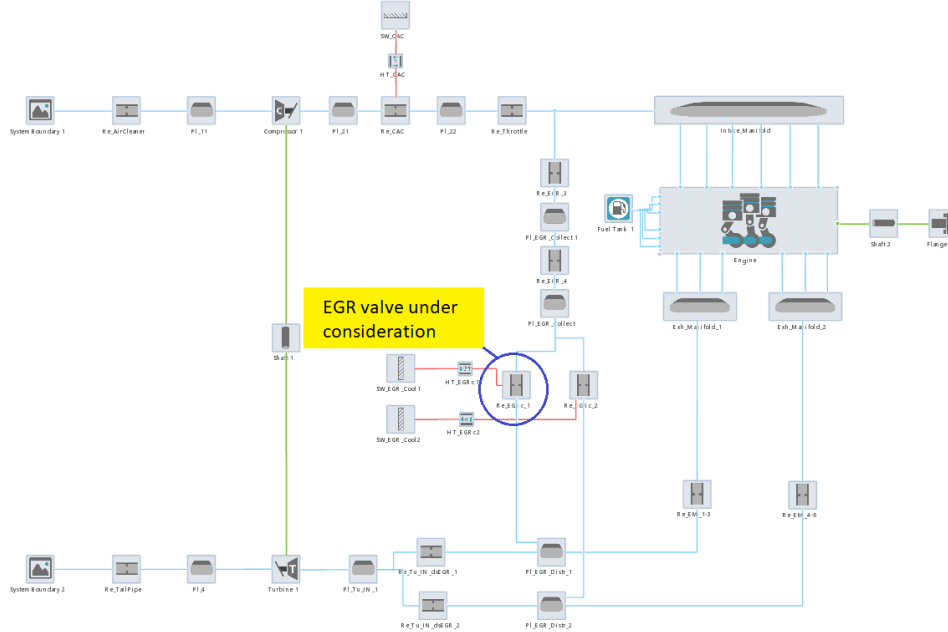


Figure 8: CruiseM model of a 13-liter, 6-cylinder, heavy duty diesel engine

An important point to be noted here is the crank angle resolved data recorded from CruiseM EGR and engine models will serve as the input in all the further simulations with the simulink model of EGR.

3 EGR Valve Model

As mentioned in the previous section, the development of EGR model is done in the software of MATLAB/Simulink. Following procedure is followed while modelling:

The mass flow through the EGR valve is calculated using the orifice equation as follows[4]

$$\dot{m} = \sqrt{\frac{2}{RT_{us}}} \psi C_d A_{geom} P_{us} \quad (1)$$

The value of ψ is given as [4]:

$$\psi = \sqrt{\frac{k}{k-1} \left[\pi^{\frac{2}{k}} - \pi^{\frac{k+1}{k}} \right]} \quad (2)$$

The ψ function in general determines the pressure ratio at which flow becomes choked. A typical behavior of ψ function can be seen in figure no.9 below

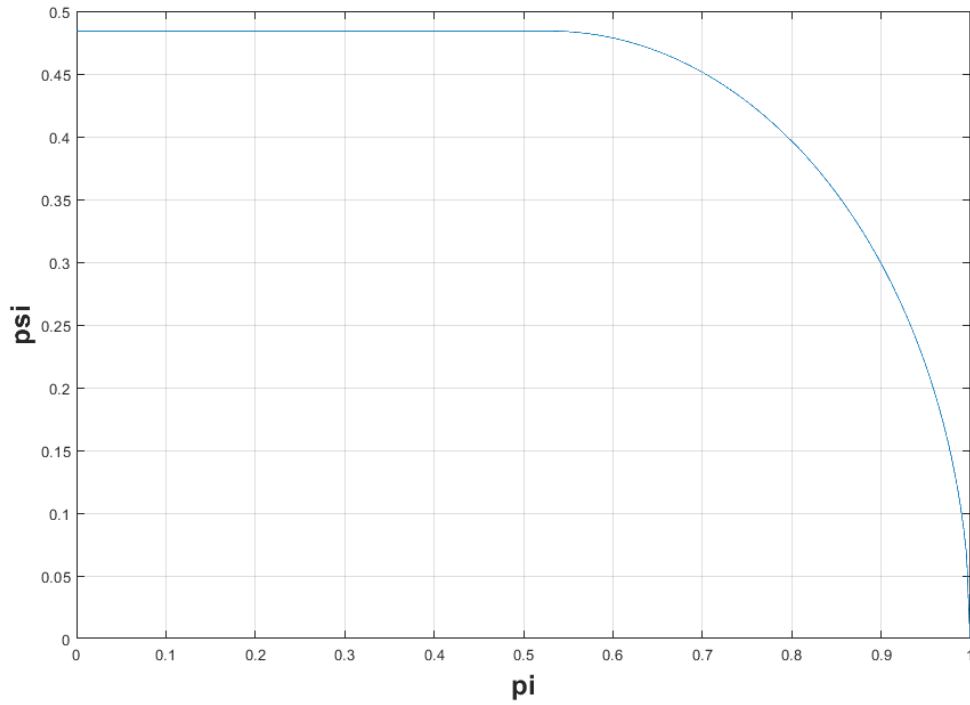


Figure 9: ψ function

The equation (2) above has an infinite slope when the pressure ratio π becomes 1 i.e. when $P_{us} = P_{ds}$, which might influence the stability of equation system. To overcome this problem, there is assumption that there exists a threshold pressure

ratio π_{tr} , after which a smooth approximation of form [4]:

$$\psi = a.(\pi - 1)^3 + b.(\pi - 1) \quad (3)$$

with

$$a = \frac{\psi'_{tr} \cdot (\pi_{tr} - 1) - \psi_{tr}}{2 \cdot (\pi_{tr} - 1)^3}, \quad b = \pi_{tr} - 3 \cdot a \cdot (\pi_{tr} - 1)^2 \quad (4)$$

is valid. ψ_{tr} is a value of ψ at the threshold pressure ratio π_{tr} , and ψ'_{tr} is the derivative of ψ_{tr} at π_{tr} [4]. The form of equation 3 guarantees finite gradient at $\pi=1$. In this thesis the threshold pressure ratio π_{tr} is considered to be 0.9952. This is because of the fact that in the reference CruiseM engine model, the same value of π_{tr} is considered. After the use of approximation equation 3 the psi function gets transformed to obtain a finite slope after the threshold pressure ratio as shown in figure no 10a and 10b below.

It has been observed that if the ψ function is modelled only with the use of approximation equation 3 (excluding equation 2), its behavior remains the same as shown in figure 10a. Hence in this thesis, only the equation 3 is used for modelling of ψ function

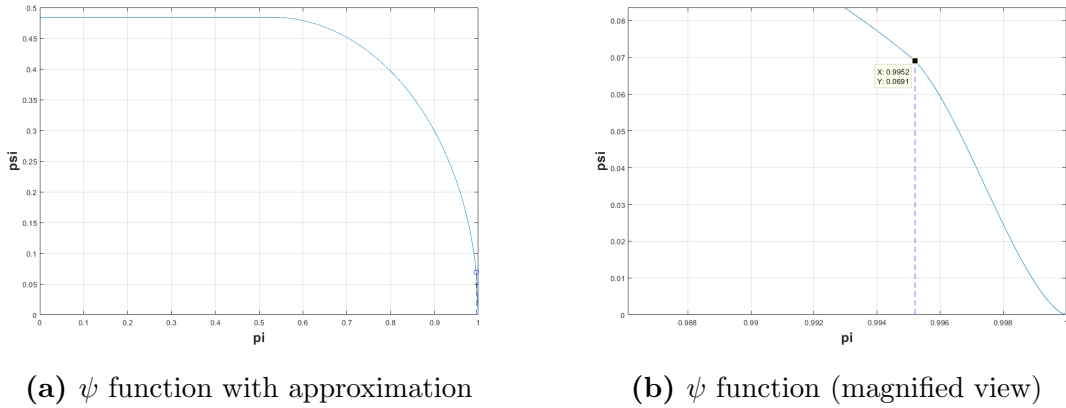


Figure 10: ψ including approximation function

The EGR model thus developed using equations above, is as shown in the figure no 14 below. This model is then simulated with the data of pressures temperatures and air-fuel ratios recorded from CruiseM EGR model shown in figure 5. The mass flow thus calculated with the simulink EGR model is then compared with the mass flow of the reference CruiseM EGR model. The comparison of the mass flows of CruiseM and simulink models is shown in the figures 11,12,and 13 below. It can be clearly seen from the figures that the mass flows of both the EGR models match vary closely. It has been found that there is less than 1% difference in the mass flows of the Simulink and CruiseM EGR models. Hence it can be easily determined that the simulink EGR model has a very good accuracy.

The details about development of each component of simulink EGR model are mentioned in sections 3.1 and 3.2

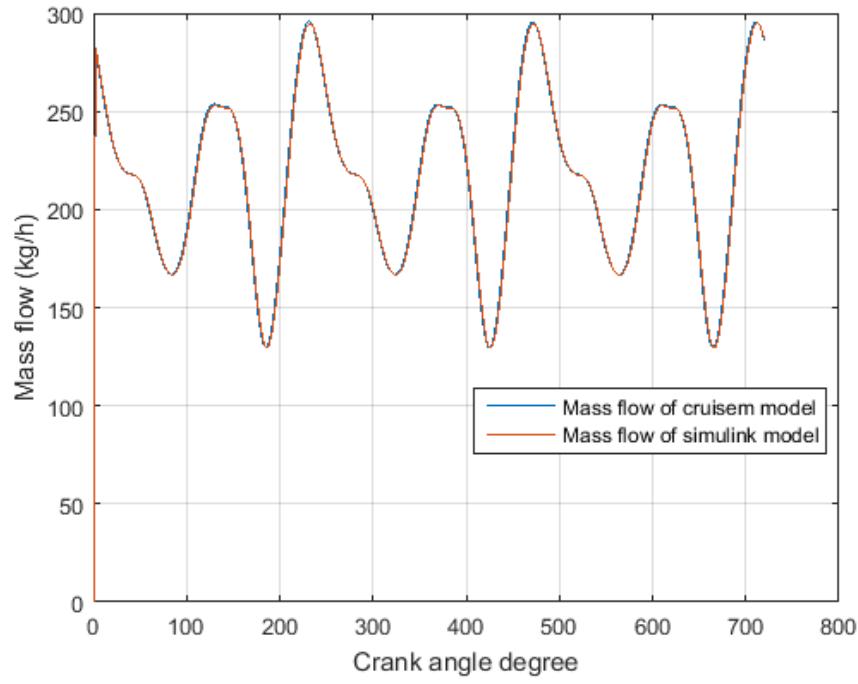


Figure 11: 100% load 1800 RPM

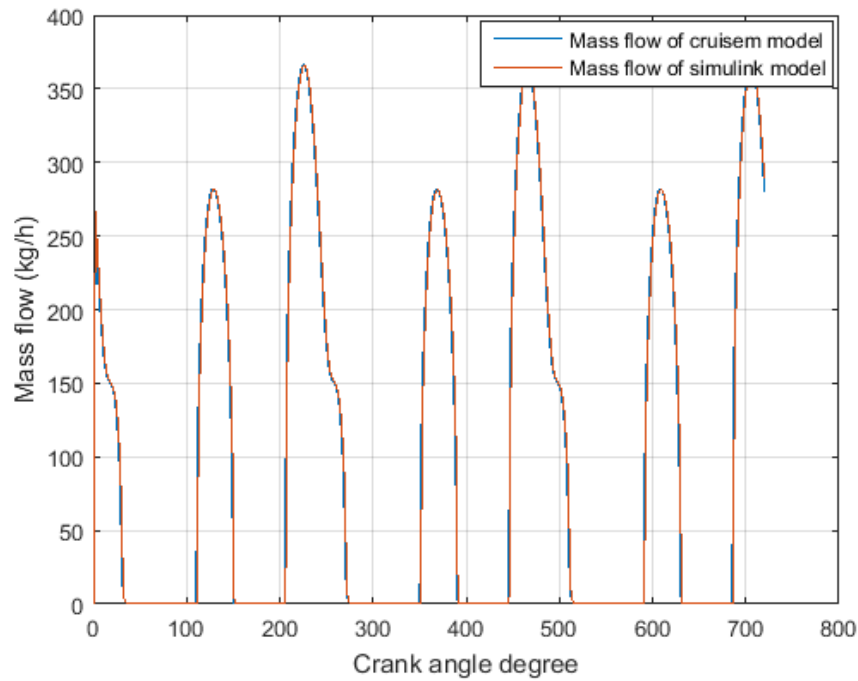


Figure 12: 100% load 1300 RPM

It can be seen from the figures 12 and 13, that at certain crank angle degrees

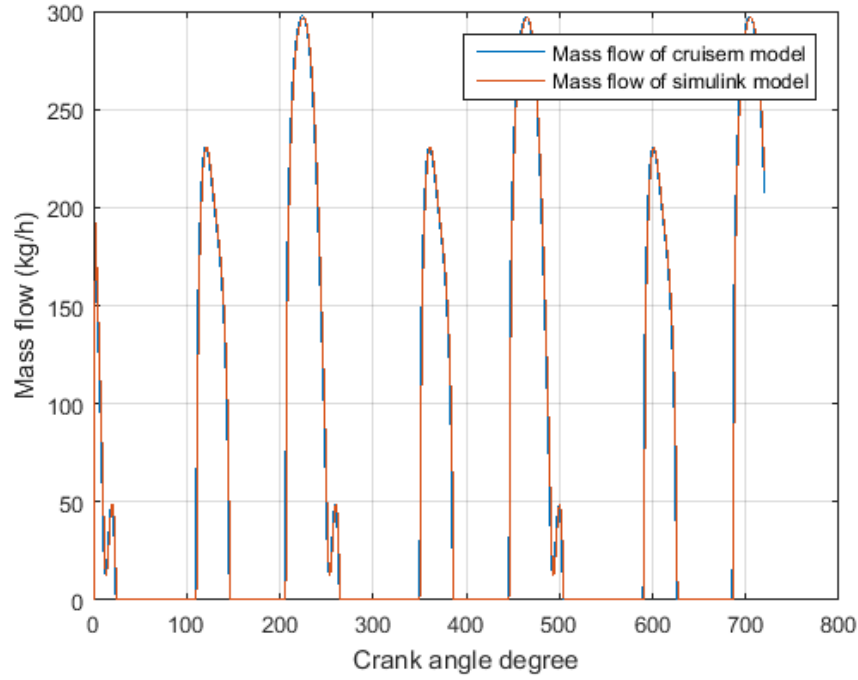


Figure 13: 100% load 1200 RPM

the mass flow calculated by both simulink and CruiseM EGR models becomes zero. The reason behind this is, at those particular crank angle degrees, the pressure ratio across the CruiseM EGR model becomes more than 1, i.e. the downstream pressure of EGR model becomes more than the upstream pressure. This situation leads to the back-flow through the EGR model. To eliminate the back-flow, the CruiseM EGR model makes the flow coefficient to be zero whenever the pressure ratio goes beyond the value of 1, thus leading to zero mass flow. In the simulink EGR model, a separate reed valve is constructed, which makes the mass flow to be zero whenever the pressure ratio goes above 1. The reed valve can be seen in the figure 14.

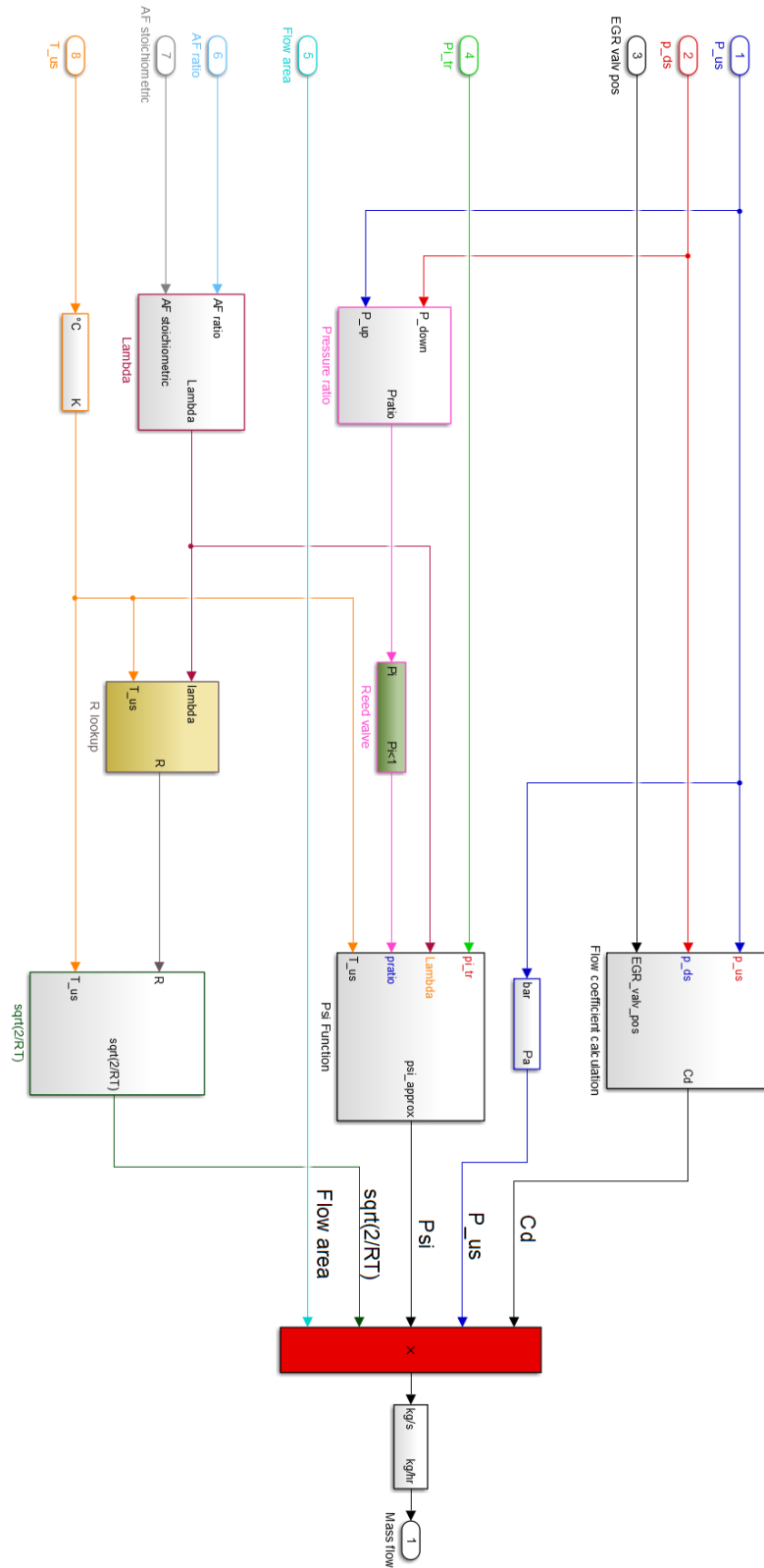


Figure 14: EGR valve model

3.1 Modelling of flow coefficient(C_d)

The flow coefficient is modelled as shown in the figure 15 below.

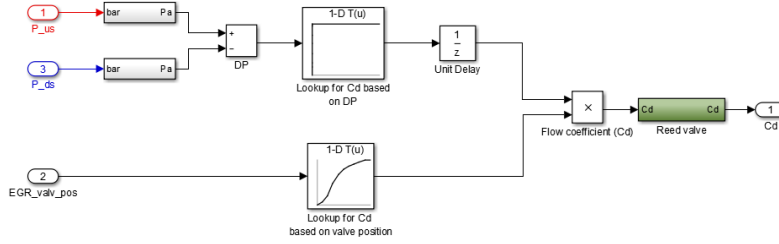


Figure 15: Model of flow coefficient

The inputs for the model of the flow coefficient are the upstream and downstream pressures and the EGR valve position. The procedure of the calculation of C_d is as follows:

1. The flow coefficient is calculated based on the differential pressure and the EGR position. The values of C_d thus calculated on the basis of above mentioned variables are then multiplied, thereby giving the final value of flow coefficient.
2. The value of the difference between upstream and downstream pressures is sent to a lookup table (shown in figure 17 below) which gives a value of flow coefficient based on the the pressure difference.
3. The EGR valve position input is given to a lookup table (shown in figure 16 below) which gives a value of flow coefficient based on the valve position.
4. The values of flow coefficient based on the differential pressure and EGR position are then multiplied, thus calculating total flow coefficient.

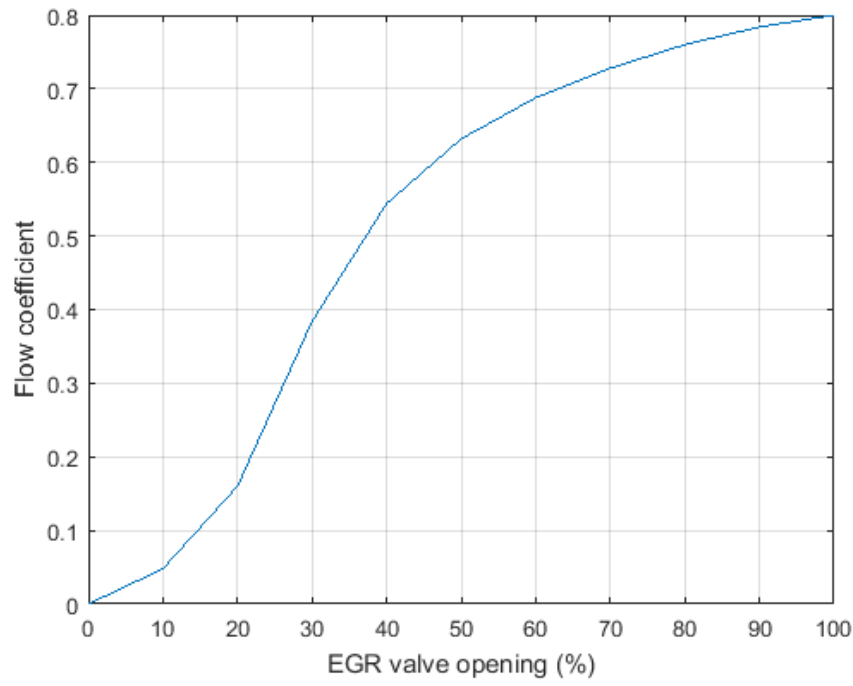


Figure 16: Flow coefficient based on EGR valve position

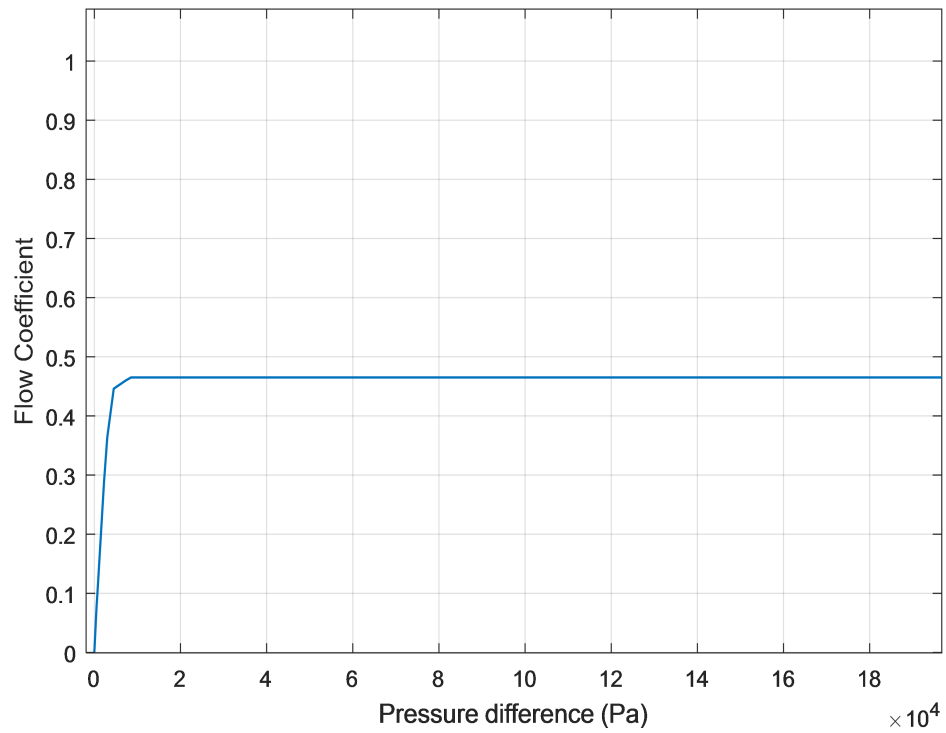


Figure 17: Flow coefficient based on pressure difference

3.2 Calculation of the gas constant (R), the ratio of specific heats (k) and the flow area (A_{geom})

R and k is determined from a lookup table which takes the input of lambda (λ) and upstream temperature (T_{us}). The values of R and k at numerous upstream temperatures and lambda values for diesel fuel are obtained from AVL CruiseM database for gases. Temperature range is considered from 200 K to 4000 K, and the lamda range is considered from 0.8 to 57.90.

Lookup tables of R and kappa plotted against lambda and temperature are as shown in figures 18 and 19 below.

The flow area A_{geom} is calculated using the equation 5 of area of cross-section for circle shown below. Where the diameter of EGR orifice (d) at throat taken from the reference is 0.044 m

$$A_{geom} = \frac{\pi}{4}(d^2) \quad (5)$$

From equation 5 above the value of flow area corresponding to the considered diameter is 0.00152053 m².

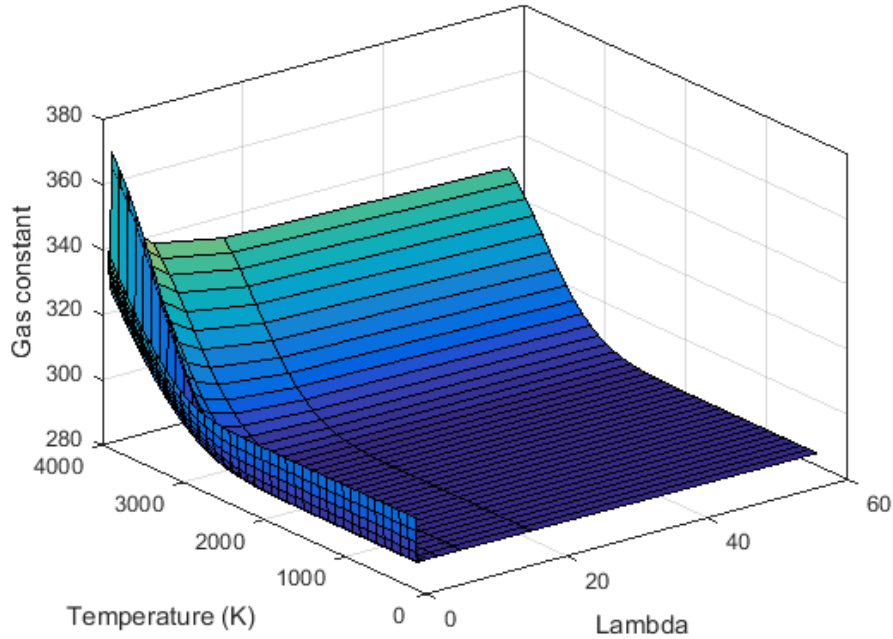


Figure 18: Gas constant(R) lookup table

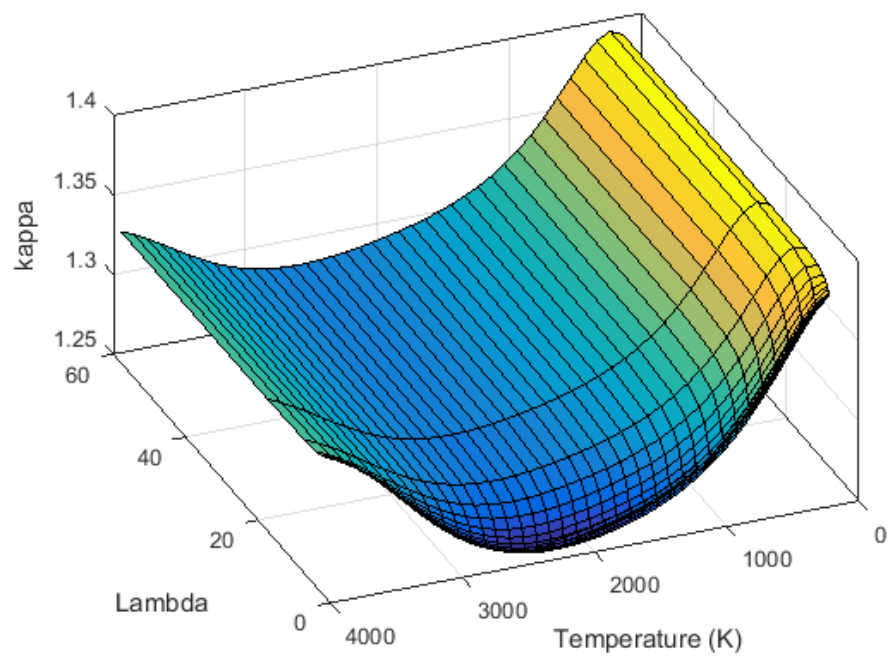


Figure 19: Kappa lookup table

3.3 Influence of π on EGR mass flow

Pressure ratio across EGR valve has a large influence on the mass flow through it. When the upstream pressure of EGR element is more than the downstream pressure (pressure ratio < 1), there will be a positive or forward flow. At certain operating points the EGR downstream pressure becomes more than that of upstream (pressure ratio > 1). When this happens the direction of flow through the EGR valve becomes reversed and there will be a backflow.

In the case when upstream and downstream pressures become equal (pressure ratio = 1) then there will not be any flow through EGR valve.



Figure 20: Upstream and downstream conditions of EGR valve

When the pressure ratio across EGR valve is reduced below the value of critical pressure ratio (pressure ratio = 0.5282), the flow through the valve will become choked. Thus, there will not be any increase in the mass flow with further decrease in the pressure ratio. The value of critical pressure ratio can be calculated using the equation 6 below.

$$\frac{P_{ds}}{P_{us}} = \left(\frac{2}{k+1} \right)^{\frac{k}{k-1}} \quad (6)$$

Following figures 21, 22 and 23 highlight the influence of pressure ratio on the mass flow through the EGR valve at various operating points. It can be seen from the figures, when the pressure ratio goes above 1, the mass flow reaches zero. The pressure ratio range of interest in this thesis is from 0.5282 i.e from the critical pressure ratio to 1.

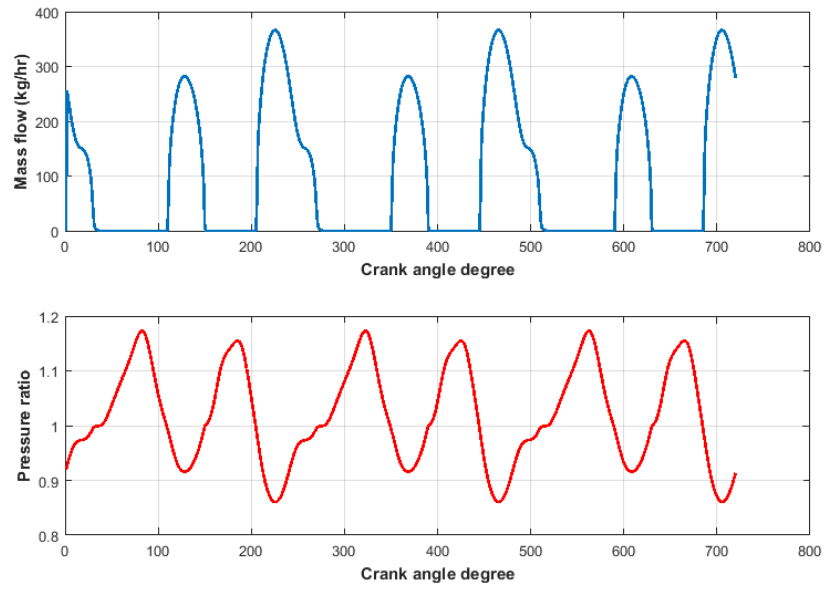


Figure 21: Effect of pressure ratio at 100% load 1300 RPM

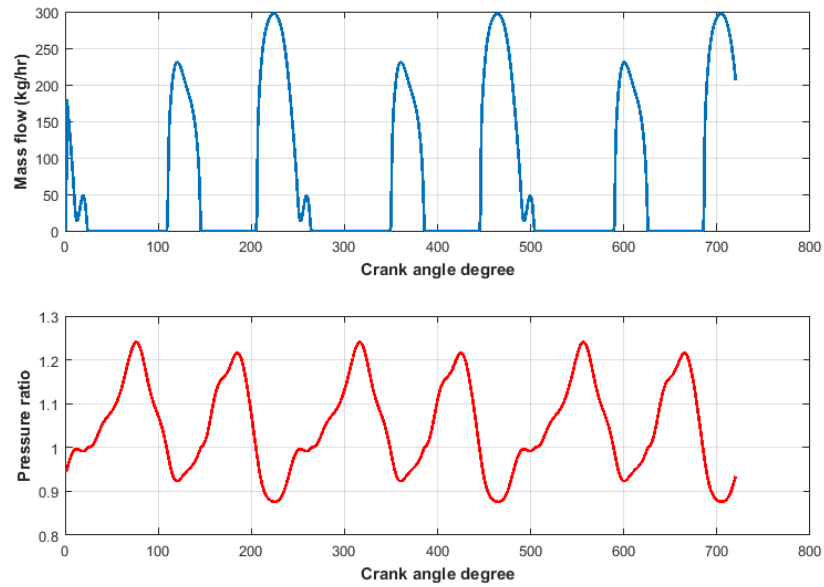


Figure 22: Effect of pressure ratio at 100% load 1200 RPM

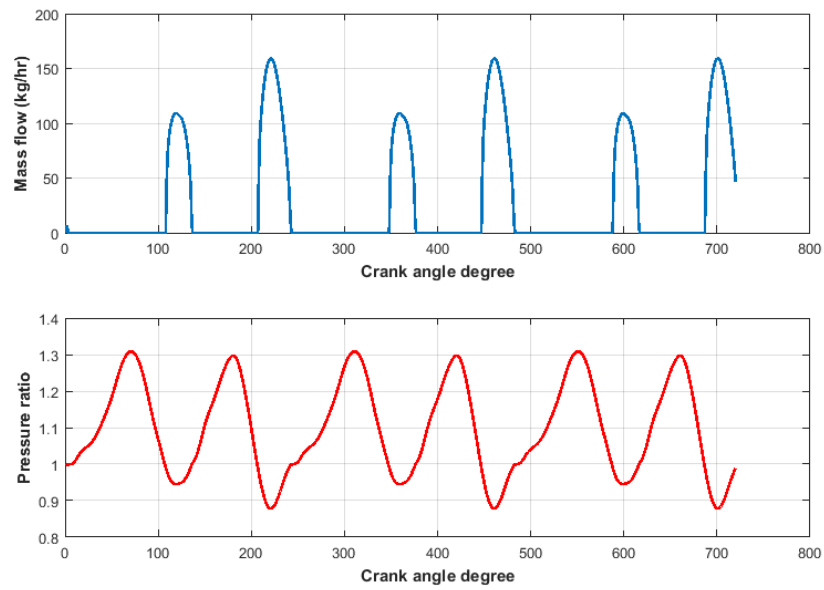


Figure 23: Effect of pressure ratio at 100% load 1000 RPM

4 Influence of constant inputs on EGR mass flow

The model in figure 14 is simulated using the crank angle resolved values of all the inputs as shown in figure 7, to calibrate it against the reference. Hence, the model is able to consider minute fluctuations in its inputs, and thus is able to calculate an accurate mass flow. However, as mentioned in section 1, a model with crank angle resolved inputs is not suitable to work real time. In order to develop a model to be real time capable, it is essential to consider mean values of inputs. However, the use of mean value models will lead to the calculation the mass flow different from the one calculated by the model with crank angle resolved inputs. Hence it is essential to analyse the effect of considering the mean value of a particular input on the mass flow through EGR model.

In this analysis, one of the inputs of EGR model is made constant (i.e mean value of the input is considered) and all other inputs are kept to be crank angle resolved. Mean mass flow thus obtained from the model is then compared with mass flow obtained from EGR model with all the inputs to be CA resolved. Following is the detailed procedure of the analysis:

1. At a particular operating point, one of the inputs to EGR model say P_{us} is kept constant. Constant here means the mean value of P_{us} is considered, which is calculated from the crank angle resolved values of P_{us} recorded from CruiseM as mentioned in section 2. While mean value for P_{us} is considered, the other inputs viz. T_{us}, P_{ds} are considered to be CA resolved. The above mentioned condition of inputs can be seen in figure 24. The figure 24 represents a block diagram of an EGR model shown in figure 14, highlighting all the inputs and the mass flow output.

It can be clearly seen in the figure that, when the mean value of P_{us} is considered, it becomes a constant input (as shown by straight line in figure 24). While the inputs other than P_{us} are considered to be CA resolved, they are shown by continuously varying waveforms. The values of threshold pressure ratio described in section 3 and flow area described in section 3.2 are also shown in the figure. It should be noted that at a particular operating point the values of EGR valve position and stoichiometric AF ratio are always constant.

2. The EGR model is then simulated with the inputs above. The mean mass flow corresponding to above inputs is then recorded.
3. This mean mass flow is then compared with the mean mass flow of an EGR model simulated with all the inputs to be CA resolved.
4. This procedure with a particular constant input is then repeated for all the operating points considered in figure 4, and comparison of mean mass flows is done.

In figure 25 the effect of considering one of the inputs of EGR model to be constant on the mean mass flow is highlighted. The figure shows the mean mass flow when the

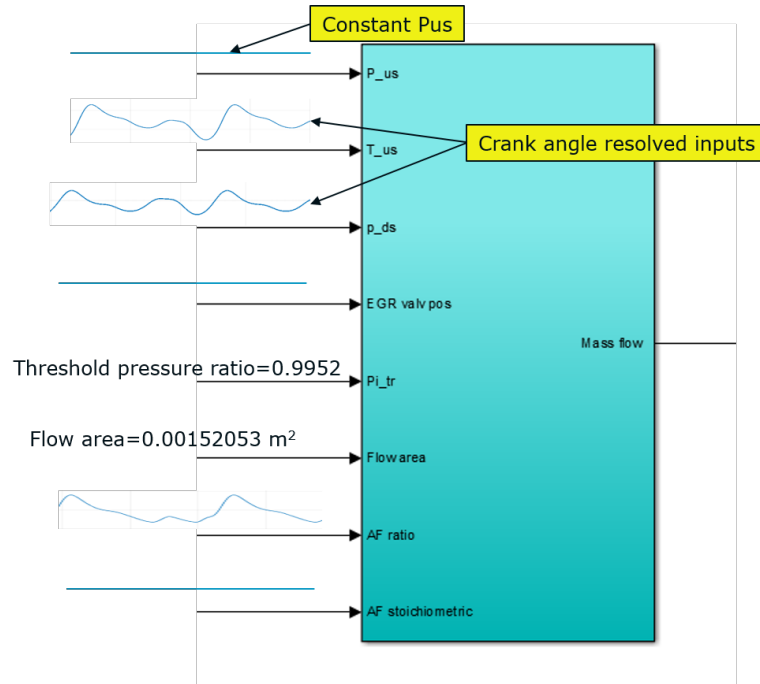


Figure 24: EGR model block diagram with P_{us} as constant

inputs P_{us} , P_{ds} , T_{us} , R , k , and C_d are kept constant simultaneously. This mean mass flow is then compared against the mean mass flow calculated by an EGR model with all the inputs to be CA resolved (reference mean mass flow). The following figure 26 shows the difference between mean mass flows of an EGR model with one of the inputs as constant and a model with all the inputs as crank angle resolved. The difference is calculated for all the operating points selected in figure 4.

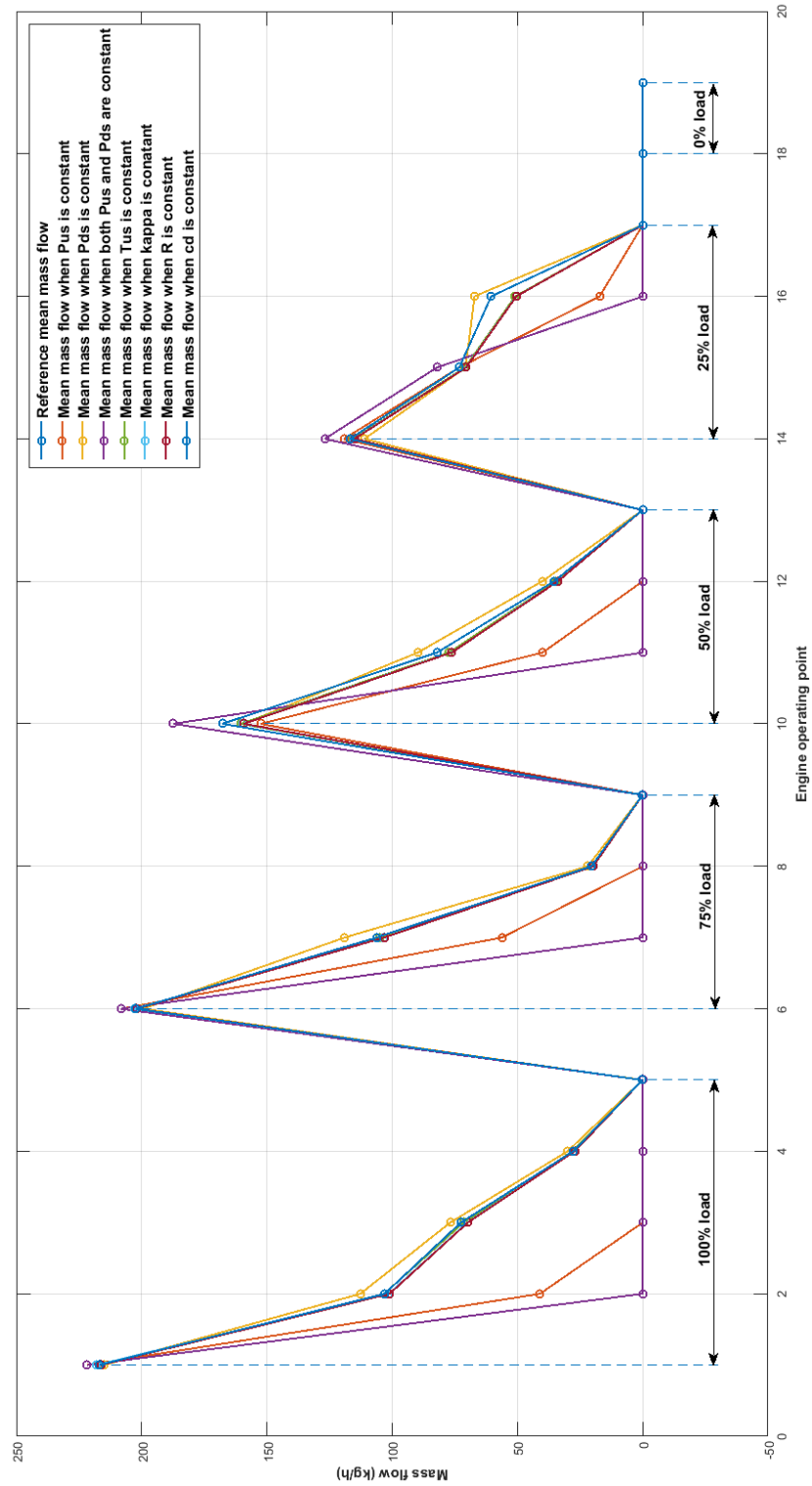


Figure 25: Comparison of mean mass flows

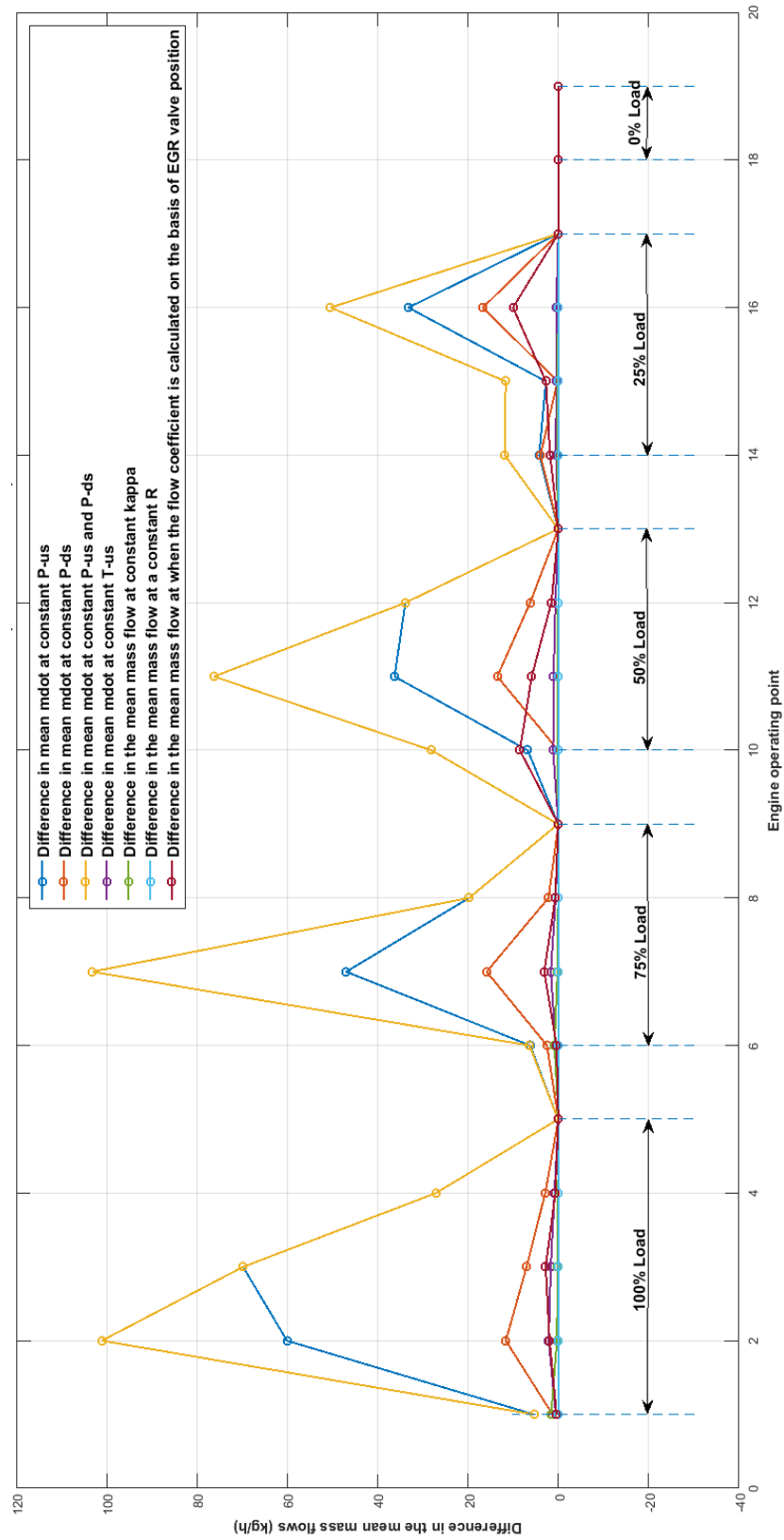


Figure 26: Comparison of difference in the mean mass flows

5 Modelling of P_{us} as a sinusoid

In further analysis the inputs to EGR model will be named as case 1, case 2 and case 3 as follows:

Case 1: EGR model with all the inputs having crank angle resolved values.

Case 2: EGR model with one of the inputs as constant, while crank angle resolved values of the other inputs have been considered

Case 3: EGR model with P_{us} as sinusoidal and mean (i.e. constant) values of all other inputs are considered.

It can be seen from figure 26, when the mean values of T_{us} , R , k and P_{ds} are considered as an input to the EGR model simultaneously, while all other inputs have been assigned with CA resolved values, the deviation in the mean mass flow calculated by case 2 EGR model is very less from the mean mass flow calculated by case 1 model. However, when the upstream pressure P_{us} is modelled as a constant input, a significant deviation between the mass flows of case 1 and case 2 models is seen.

As mentioned in section 1, one of the requirements of the real time mean-value EGR model is its mean mass flow should be the same as mean mass flow calculated by case 1 model. Since the mean value models can't consider variations occurring in the inputs per engine cycle, they are unable to calculate accurate mean mass flow.

One of the ways to reduce the deviation between EGR mass flow calculated by a mean value EGR model and a case 1 model, is to model P_{us} as a sinusoidal signal, while modelling other inputs as constant (i.e considering their mean values). The reason behind that is P_{us} has a large effect on a mass flow through EGR element as compared to any other inputs, as highlighted by figure 26. Thus, P_{us} is modelled using the equation of sine wave as below:

$$P_{us_sine} = A \cdot \sin(2\pi ft) + P_{mean} \quad (7)$$

Following procedure is followed while modelling sinusoidal P_{us} :

1. At a particular operating point the maximum (P_{max}) and mean values of P_{us} (i.e. P_{mean}) are calculated from the crank angle resolved values of P_{us} recorded from CruiseM.
2. The frequency f of the sinusoidal P_{us} shown in equation 7 is calculated from the RPM of the particular operating point as follows:

At a particular RPM say 1800 RPM, the rotations per second are 30. The reference CruiseM engine in this thesis has 6 cylinders. Hence for a 6 cylinder engine there are 3 pulses of exhaust flow every rotation. Thus the number of exhaust pulses per second are 90, therefore the frequency becomes $f = 90$ Hz.

General formula for f is thus given as:

$$f = (RPS) \times 3 \quad (8)$$

3. The amplitude (A) is calculated by taking the difference of P_{mean} from the maximum value of P_{us} .

$$A = P_{max} - P_{mean} \quad (9)$$

Thus, after all the above calculations the sinusoidal upstream pressure P_{us_sine} for the operating point of 100% load and 1800 rpm can be seen as in figure 27 below:

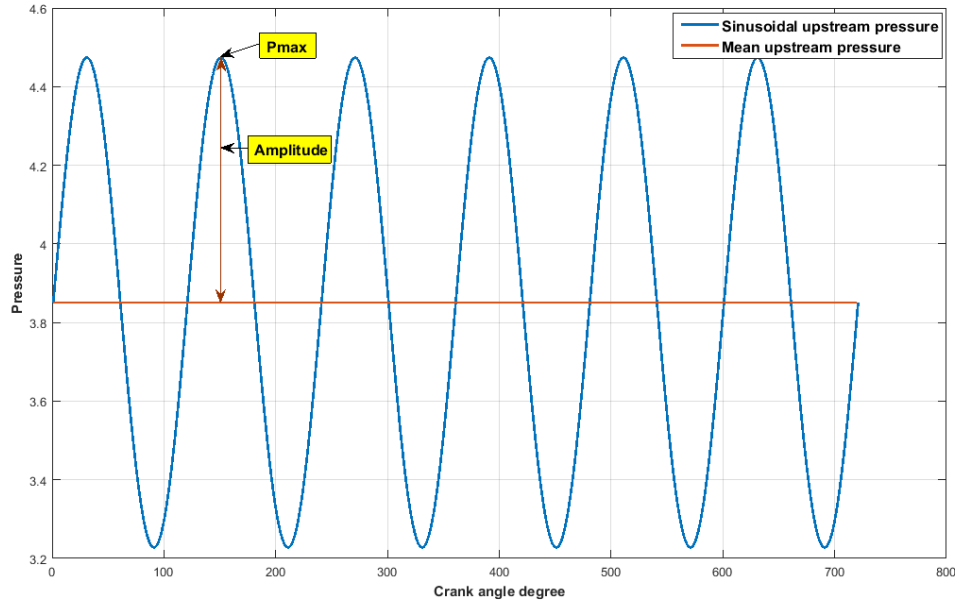


Figure 27: Sinusoidal upstream pressure

A block model of an EGR model with case 3 input conditions is as shown in figure 28. It can be seen from the figure that, the inputs of AF ratios used for the calculation of R and k are eliminated and constant values of $R = 287 \text{ J/kg.K}$ and $k = 1.4$ are considered instead (the value of R is shown in 28, while the value of k is not shown). The internal structure of the block diagram 28 with case 3 inputs, is shown in figure 30.

Figure 29 shows the comparison of mean mass flows calculated by an EGR model with case 3 inputs modelled as shown in figure 30 and by the EGR model with case 1 inputs modelled in figure 14. The comparison is done for the operating points shown in figure 4.

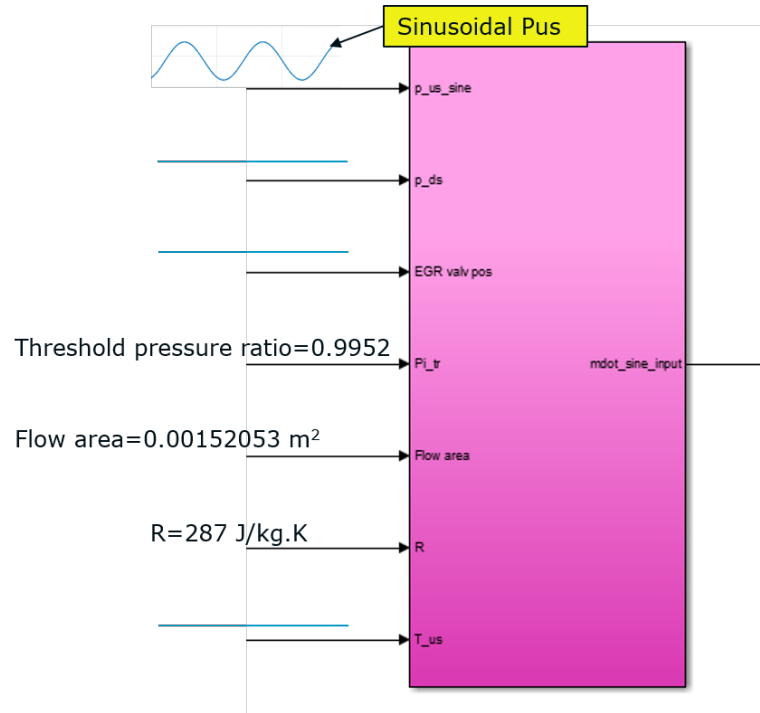


Figure 28: EGR block diagram highlighting the inputs

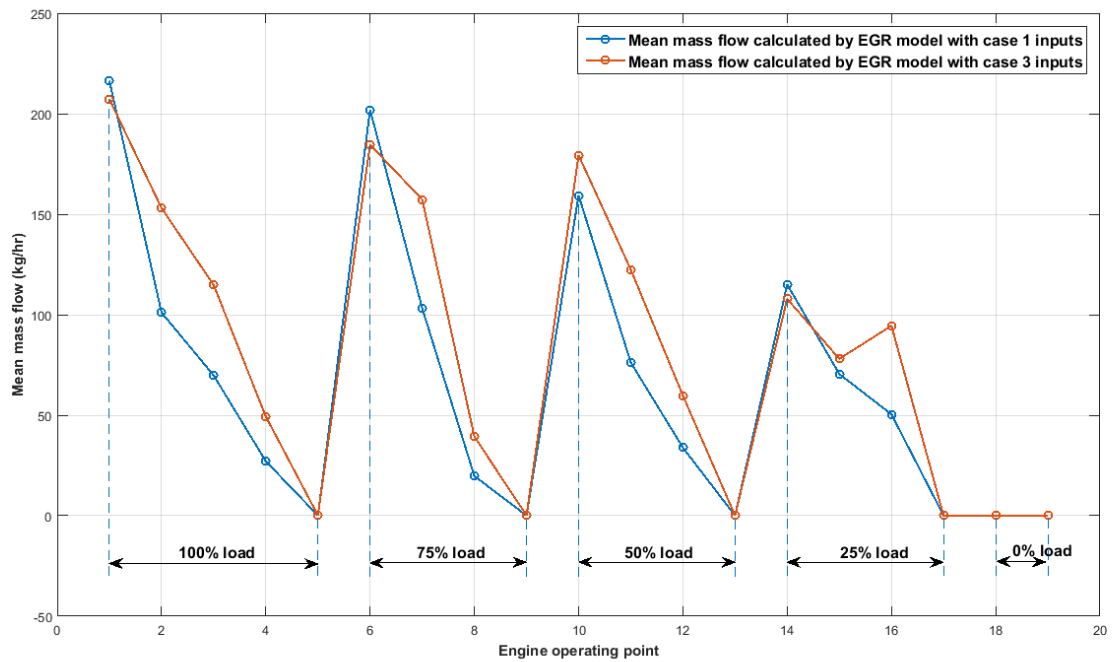


Figure 29: Comparison of mean mass flows calculated by EGR models with case 1 and case 3 inputs

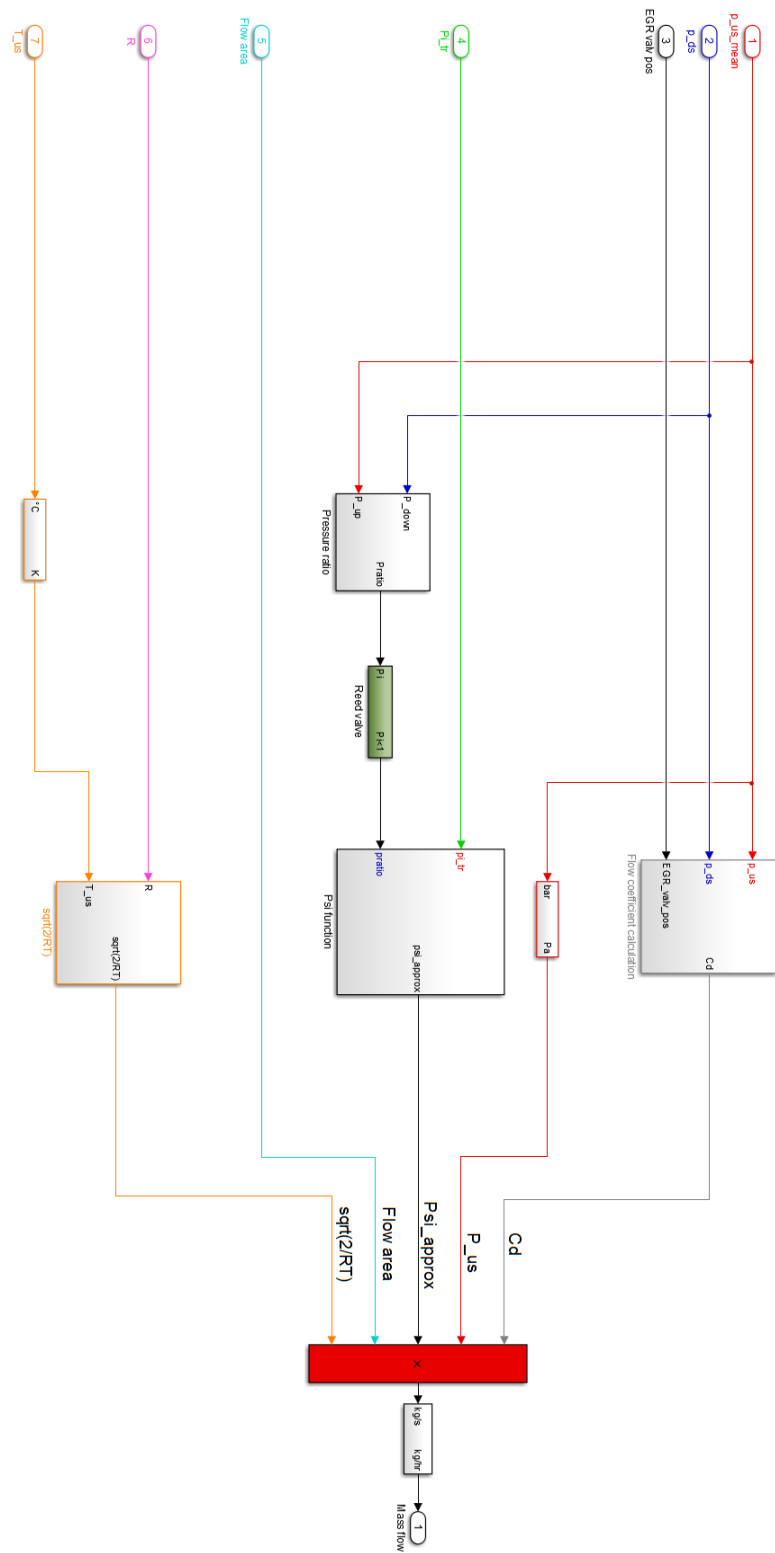


Figure 30: EGR model with P_{us} as sinusoidal and all other inputs to be constant

5.1 Consideration of harmonics

It can be seen from the figure 29 that for most of the operating points, the mean mass flow calculated by an EGR model with case 3 inputs, varies largely from the mean mass flow calculated by EGR model with case 1 inputs. One of the ways to compensate for this difference is to incorporate frequency content (harmonics) into the sinusoidal P_{us} input of EGR model shown in figure 30. With the addition of harmonics in sinusoidal P_{us} it is possible to recreate the crank angle resolved waveform of the P_{us} (for example as shown in figure 7a) which is recorded from CruiseM at a particular operating point. If the shape of the waveform of P_{us} resembles to its CA resolved waveform then there is good possibility that the difference between the mass flows of EGR models with case 3 and case 1 inputs can be reduced. Following procedure if followed to incorporate harmonics in the sinusoidal P_{us} :

1. At a particular operating point as mentioned in figure 4, the Fast Fourier Transform of the CA resolved waveform of P_{us} is taken thereby revealing the frequency content in the waveform. A CA resolved waveform of P_{us} at the operating point of 100% load and 1800 rpm and its frequency content is shown in figure 31. As can be seen from the figure that there are total 3 dominant harmonics. Same amount of governing harmonics have been found in the P_{us} of all the selected operating points. Hence it is decided to consider maximum 3 harmonics in the sinusoidal P_{us} simultaneously.

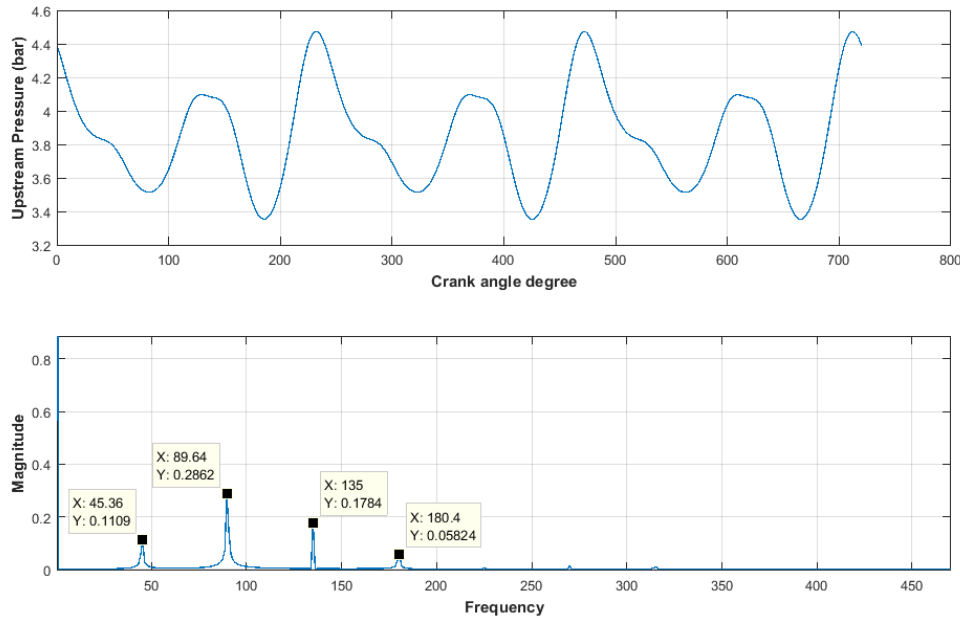


Figure 31: Frequency content of P_{us} input at 100% load and 1800 RPM

- The harmonics are added to the sinusoidal waveform of the upstream pressure P_{us} . The figure below highlights the change in the sinusoidal waveform of P_{us} with the addition of harmonics.

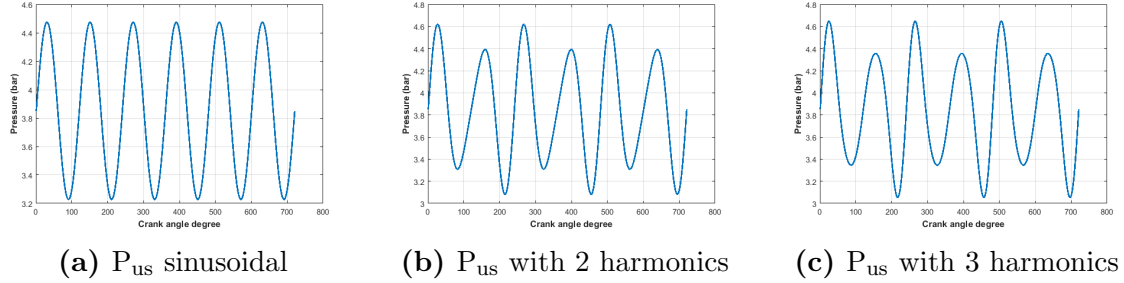


Figure 32: Change in the waveform of P_{us} with the addition of harmonics

- Thus with addition of harmonics in the sinusoidal waveform of P_{us} while keeping all other inputs of EGR model to be constant (i.e. considering their mean values) the EGR model shown in figure 30 is simulated at various operating points chosen in figure 4. The mass flow thus obtained from the model is compared with the mass flow of EGR model with case 1 inputs.

Following figures 33 and 34 compare the mass flows of EGR model with case 3 inputs including the consideration of harmonics in P_{us} and the mass flows of EGR model with case 1 inputs.

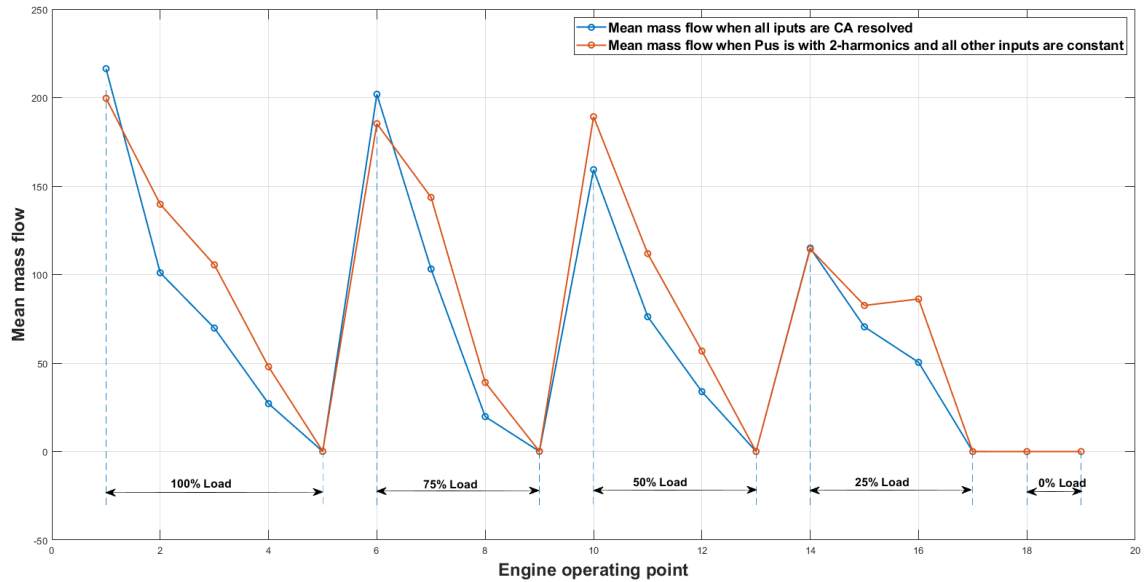


Figure 33: Comparison of EGR mean mass flows with case 3 inputs including 2 harmonics in P_{us} and case 1 inputs

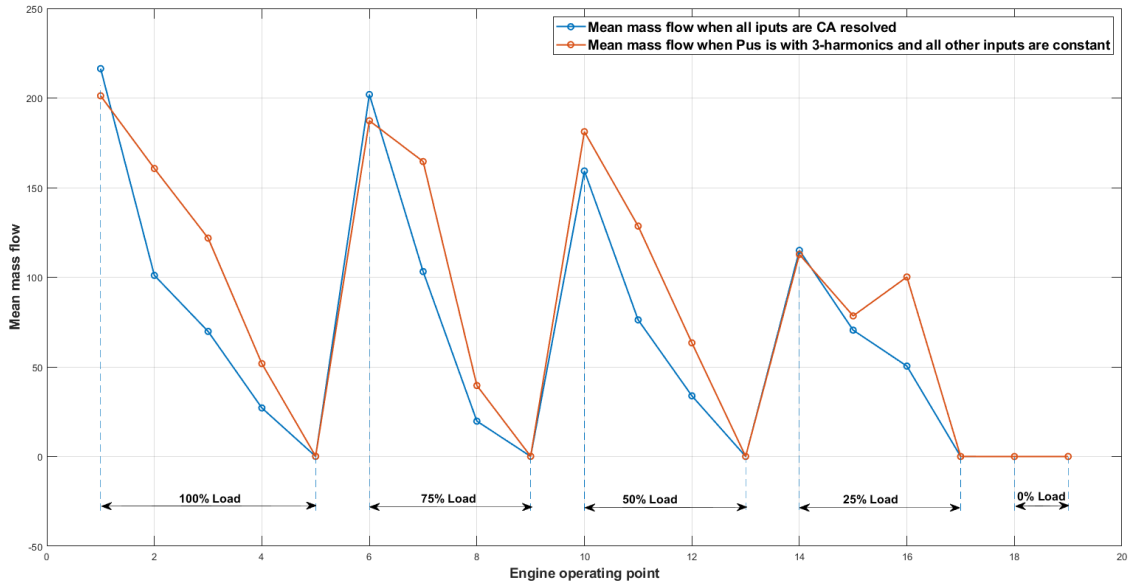


Figure 34: Comparison of EGR mean mass flows with case 3 inputs including 3 harmonics in P_{us} and case 1 inputs

It can be seen from the figures 33 and 34 that even with the consideration of harmonics, there is significant difference between the mass flows calculated by EGR models with case 1 and case 3 inputs. Hence, the proposition of the addition of harmonics with sinusoidal upstream pressure was not extended further.

5.2 Influence of amplitude on the mass flow

In section 5 an EGR model with case 3 inputs has been developed. In this model the amplitude of sinusoidal P_{us} is calculated using equation 9. The amplitudes calculated with above equation causes some contrast between mean mass flows calculated using models with case 1 and case 3 inputs, as highlighted by fig. 29. It has been discovered in this thesis, that the amplitude of P_{us} has a significant influence on the mass flow calculated by an EGR model with case 3 inputs. It has also been noted that one of the ways to achieve similar mean mass flow as that of reference, using the EGR model with case 3 inputs is to consider suitable amplitudes for modelling of P_{us} as a sinusoidal. In order to find suitable amplitudes, a trial and error method is used in this thesis. The procedure is explained in more details as below.

1. At a certain operating point chosen in figure 4, firstly an EGR model shown in figure 30 is simulated with case 3 inputs and the mean mass flow thus obtained is compared with the mean mass flow of EGR model with case 1 inputs as shown in fig 14.
2. If the mean mass flows are not same then the amplitude of sinusoidal P_{us} is manipulated i.e. increased or decreased, without any change in the remaining

inputs at that particular operating point.

3. For each change in the amplitude the corresponding change in the mass flow is tracked. The value of amplitude for which the mean mass flows of EGR models with case 1 and case 3 inputs become the same is chosen as the suitable value of amplitude. Thus for each operating point mentioned in the figure 4 the suitable value of amplitude is determined.

The figure 35 below shows the amplitudes for P_{us} calculated using eqn. 9 and the amplitudes determined using above mentioned trial and error method. The figure 36 highlights the fact that the mean mass flow calculated using the EGR model with case 3 inputs can be perfectly matched with the mean mass flow of EGR model with case 2, if the amplitude of P_{us} is determined using the trial and error method mentioned above.

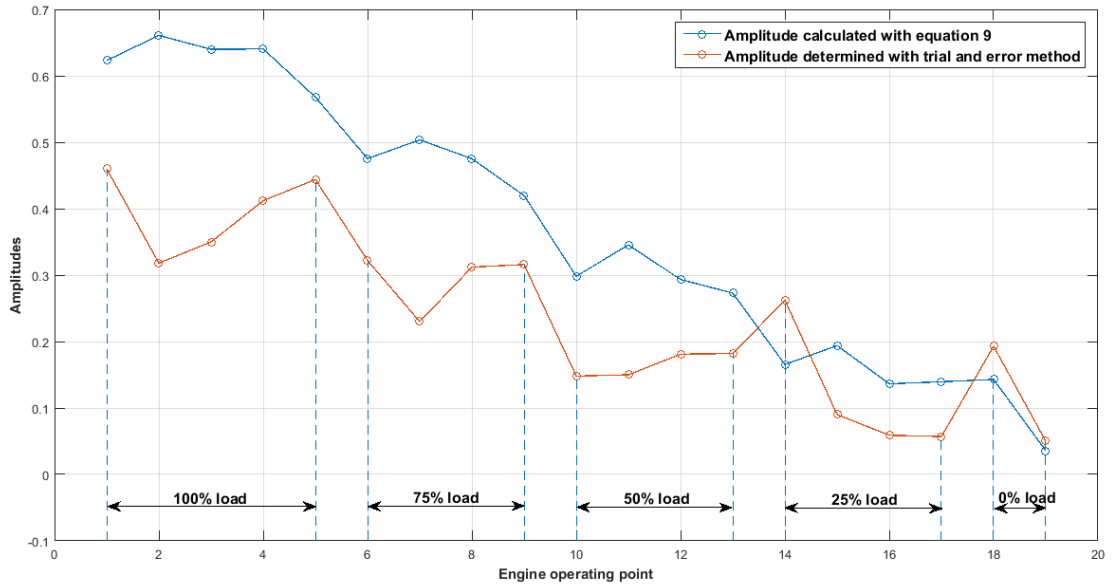


Figure 35: Comparison of actual and required amplitudes

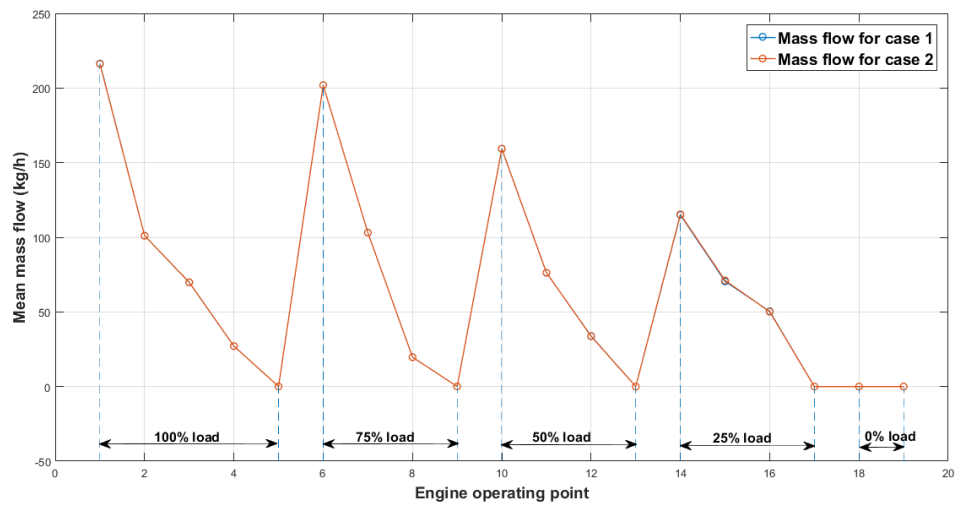


Figure 36: Comparison of mean mass flows of case 1 and case 3

6 Real time capable EGR model

It can be seen from the above sections, that a mean mass flow calculated from an EGR model with case 1 inputs (figure no 14) can be emulated by a model with case 3 inputs (figure no 30). The EGR model developed with case 3 inputs has a faster calculation speed than the one with case 1 inputs, but still it is not a real time capable and hence it cannot be implemented in an ECU.

The EGR model with case 3 inputs need the upstream pressure to be modelled as a sinus, in order to calculate the mass flow. If the model is implemented in ECU, then during actual engine operation the generation of sinusoidal P_{us} will consume significant computational effort of ECU, thereby making the calculation of mass flow to be slower.

One way to improve the calculation speed of the model is to eliminate the necessity of the model to generate the sinusoidal P_{us} . The EGR model described in this section eliminates the necessity to generate the sinusoidal upstream pressure, but instead it calculates the mass flow by taking the mean upstream pressure and its amplitude as separate inputs. Thus, the model is just a modification in the EGR model with case 3 inputs. Since the model eliminates the necessity to model the upstream pressure as a sinusoid, it is able to calculate the mass flow in real time, thus making it suitable for its implementation in ECU. The block diagram and actual EGR model are shown in the figures 37 and 38 below.

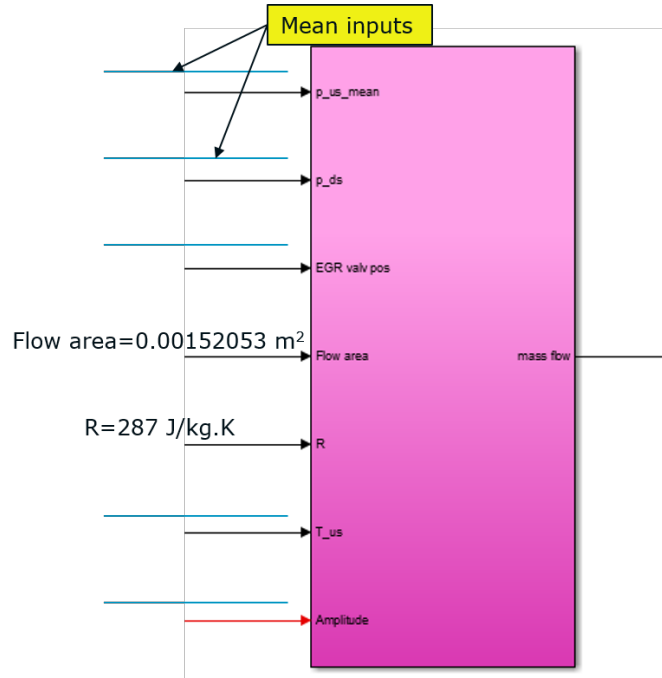


Figure 37: EGR real time capable block model

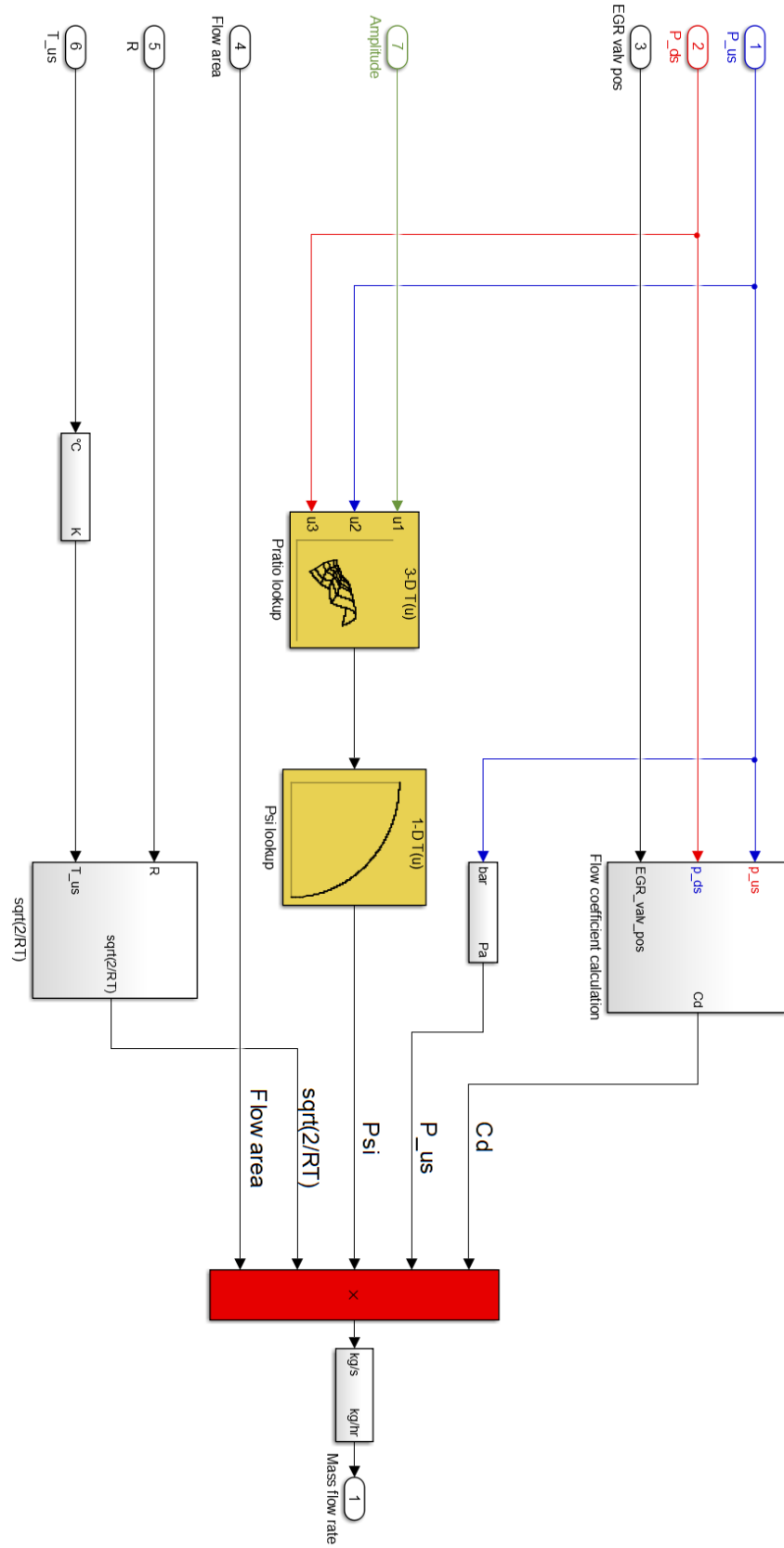


Figure 38: EGR valve real time model

It can be seen from the figure 37 that there is additional input of 'Amplitude'. This amplitude corresponds to sinusoidal P_{us} and is calculated using equation 9. It can also be noted that the mean values for all the inputs are considered in this model, as highlighted by straight lines in figure 37. As mentioned in the previous sections, the mean values of all the inputs are calculated from the crank angle resolved data of inputs, recorded from the AVL CruiseM model at various operating points.

The important components of the real time EGR model shown in figure 38 are as below

1. Lookup table for pressure ratio.
2. Lookup table for Psi function.

6.1 Lookup table for pressure ratio

This lookup table is a 3 dimensional, and its inputs are as follows:

1. Amplitude calculated using eqn.9.
2. Mean P_{us} and mean P_{ds} .

The output of the lookup table is pressure ratio. In the model shown in figure 38, instead of modelling P_{us} as a sinusoidal signal, the amplitude and the mean value of P_{us} are considered as a separate inputs to the 3d lookup table.

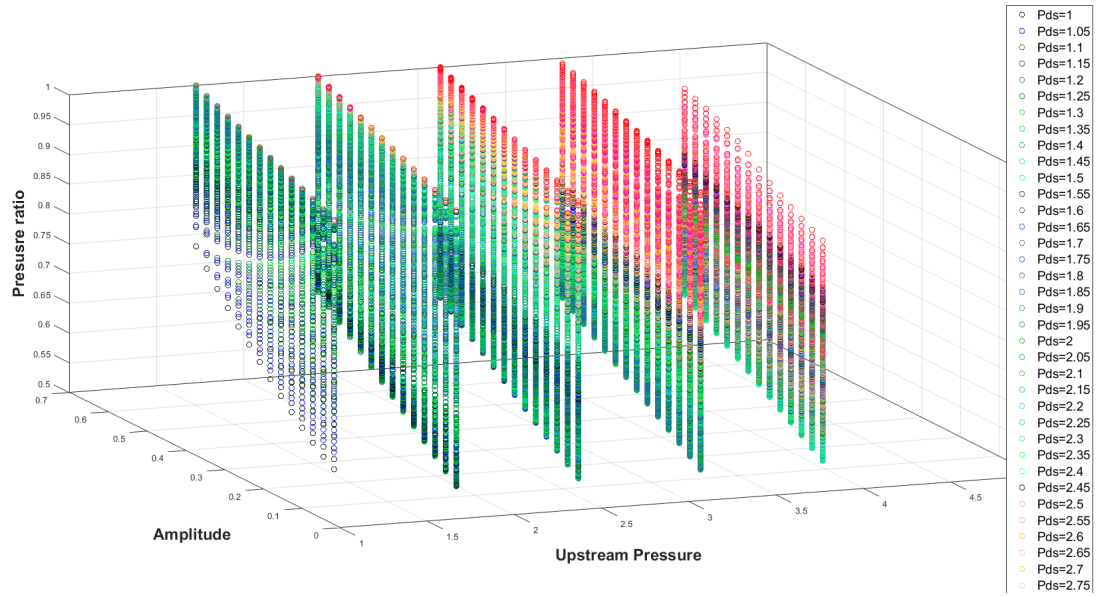
The procedure below is followed to generate the 3d lookup table:

1. An EGR model with case 3 inputs (model shown in figure 30) is used for generating the lookup table. An important requirement for the lookup table is that it should be globally valid, i.e. the table must be applicable to every engine and not just the reference engine model used in this thesis.
2. Firstly, an EGR model with case 3 inputs (e.g. the model shown in figure 28) is simulated with the range of inputs shown in table 1 and the mean mass flow for each combination of inputs is recorded. It can be seen from the table the ranges of inputs chosen for simulation involves maximum and minimum pressures and the pressure amplitudes occurring in most of the commercial engines. It should also be noted that the value of flow coefficient is considered to be 1. A constant value of temperature (650°C) is assumed for all the ranges of pressures and amplitudes mentioned in the table. The reason behind the consideration of constant temperature will be explained in further description.
3. The mass flow recorded from the above mentioned simulation is then given to an inverse model of EGR valve as shown in figure 43. The inverse model primarily takes the input of mass flow as calculated from the model mentioned in above description, and divides it with the product of flow area, mean P_{us} and the term $\sqrt{RT_{us}}$, thus calculating the value of ψ .

Input	Range	Number of points
Amplitude of P_{us}	0.05:0.05:0.7	14
Mean value of P_{us}	1.077:0.05:4.527	70
Mean value of P_{ds}	1:0.05:3.5	51

Table 1: Range of the inputs of the lookup table

4. The ψ value is then provided to a 1D lookup table which determines the value of pressure ratio corresponding to a particular ψ value.
5. The values of pressure ratios thus obtained are then used to populate a 3D lookup table. The 3D lookup table populated with the values of pressure ratios is as shown in figure 39.

**Figure 39:** 3D lookup table

In this figure, on X and Y axis there are upstream pressure P_{us} and Amplitude respectively, and on Z axis the pressure ratio is shown. The color codes represent downstream pressure P_{ds} . The lookup table contains total 49980 values of pressure ratio at different combinations of inputs amplitude, P_{us} and P_{ds} shown in table 1.

In order to show the detailed behavior of the pressure ratio, the following figures 40 and 41 are included. The figures represent the behavior of pressure ratio at the downstream pressure of 1 bar and 1.5 bar.

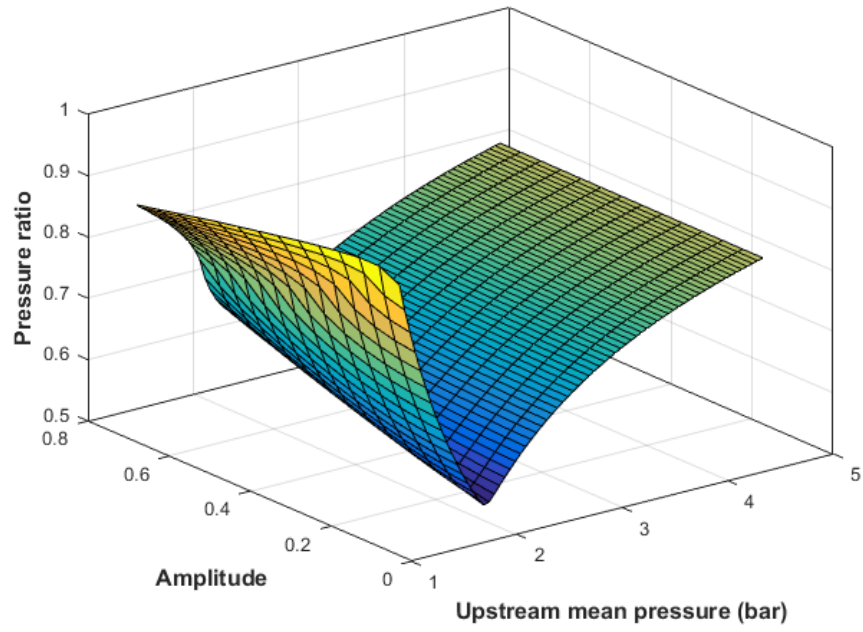


Figure 40: Variation of pressure ratio at $P_{ds}=1$ bar

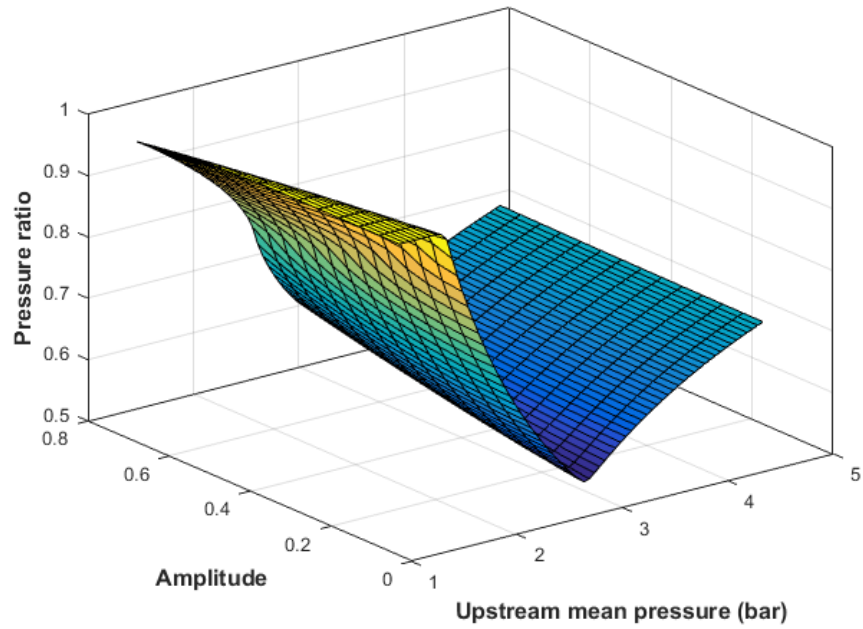


Figure 41: Variation of pressure ratio at $P_{ds}=1.5$ bar

6. It has been observed that the temperature has no effect on the determination of pressure ratio. This effect can be explained as follows:

- At a particular combination of inputs mentioned in table 1 and at the temperature of 650°C, certain value of mean mass flow is calculated from an EGR model with case 3 inputs as mentioned in bullet point 2 above. The pressure ratio corresponding to the mean mass flow will be calculated from the inverse EGR model as described in bullet point 3 .
- When the value of temperature is changed, let's say to 350°C, while no change is made in the values of other inputs (viz pressures, and amplitude) there will be change in the mass flow corresponding to the change in temperature.
- However when this mass flow is given to the inverse model for the calculation of pressure ratio, the effect of change in mass flow due to the temperature is nullified because of term $\sqrt{RT_{us}}$. Thus the value of pressure ratio calculated at the temperature of 350°C will be the same as the one calculated at 650°C.

6.2 Lookup table for Psi (ψ) function

This lookup table is a 1D lookup which takes the input of pressure ratio determined by the 3D lookup table described in above section and conveys the value of ψ . The ψ value for the lookup table is calculated using equation 3 for the pressure ratio range of interest mentioned in section 5. Following figure shows the plot of the 1D lookup table for ψ function.

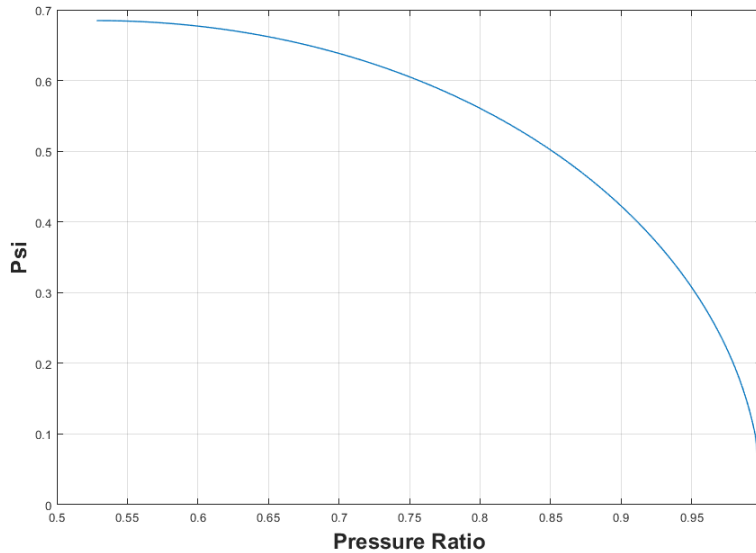


Figure 42: Psi lookup table

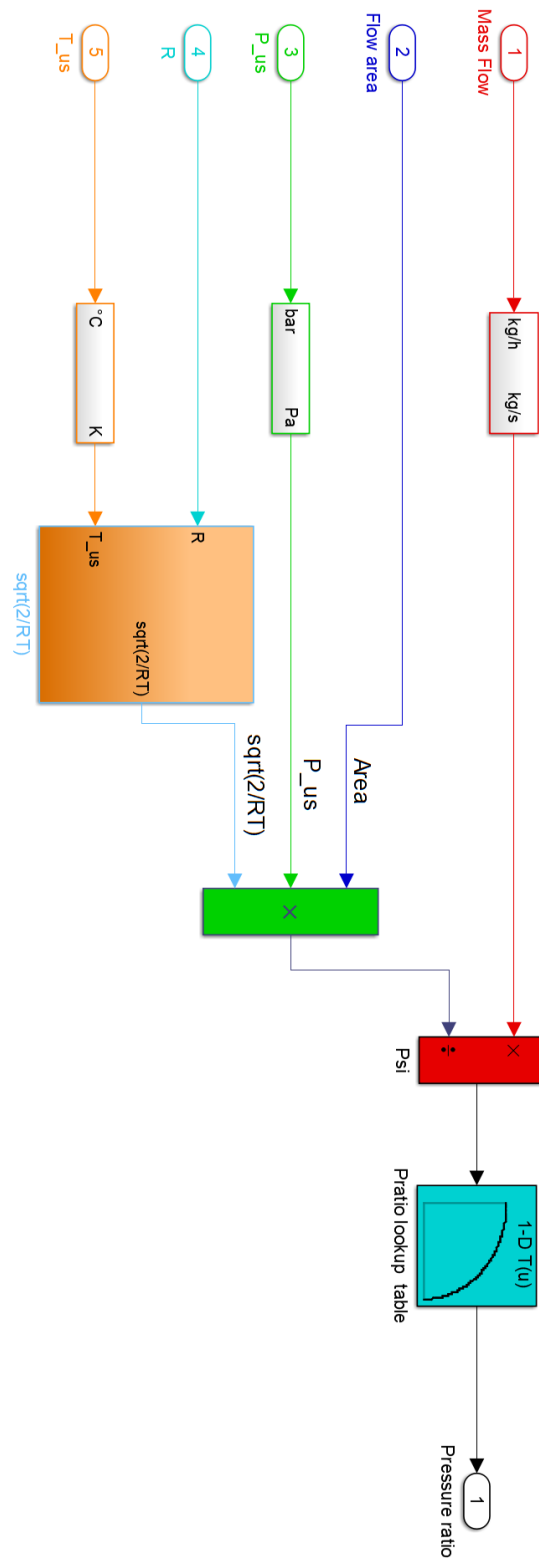


Figure 43: Inverse model

6.3 Results of real time model

The mean mass flows calculated by the real time model developed in above sections are compared against the mean mass flows of EGR model described in section 3 i.e. EGR model with case 1 inputs. Both EGR models are simulated at all the selected operating points mentioned in figure 4 with the input data recorded from CruiseM EGR and engine model. Important point to be noted here is, in the real time model, the amplitudes used to describe the sinusoidal behavior of P_{us} are the ones determined using trial and error method as described by figure 35. The mean mass flows thus obtained from the simulation of EGR models are then compared. The figures below show the comparison of the mean mass flows calculated by both the EGR models described above.

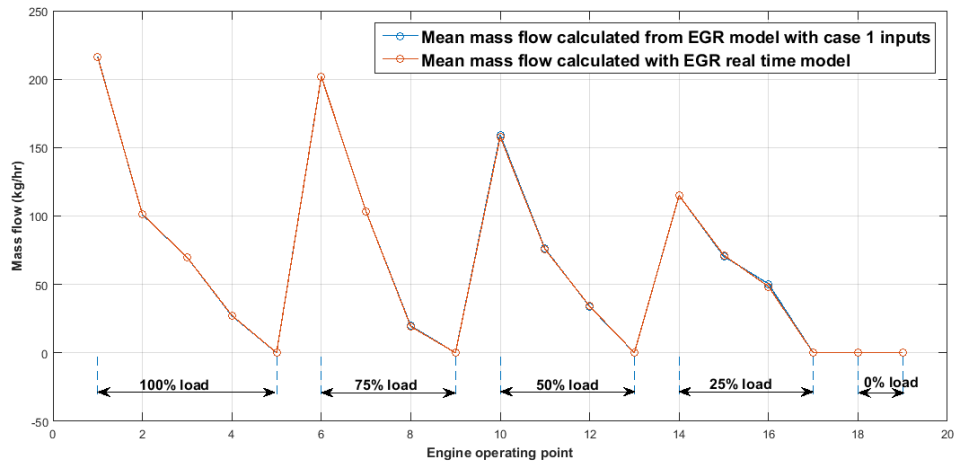


Figure 44: Comparison of the mass flows calculated with EGR model having case 1 inputs and real time EGR model

It can be clearly seen from the above figures 44, 45 and 46, that the mean mass flows of the real time model shown in figure 38 and EGR model with case 1 inputs (shown in figure 14) match very closely. Figure 45 highlights that maximum difference of 2.02 kg/hr occurs between the mean mass flows. As mentioned before the EGR model developed in figure 14, has all the input values to be crank angle resolved. Thus the model is able to consider the variations occurring in the pressures and temperatures in each engine cycle, and thus calculate the accurate mass of gases going towards the intake manifold. Since the mean mass flow of the real time model matches closely with the mean mass flow calculated by the model in figure 14, it can be said that the real time model (shown in fig. 38) has a good accuracy in the calculation of mass flow.

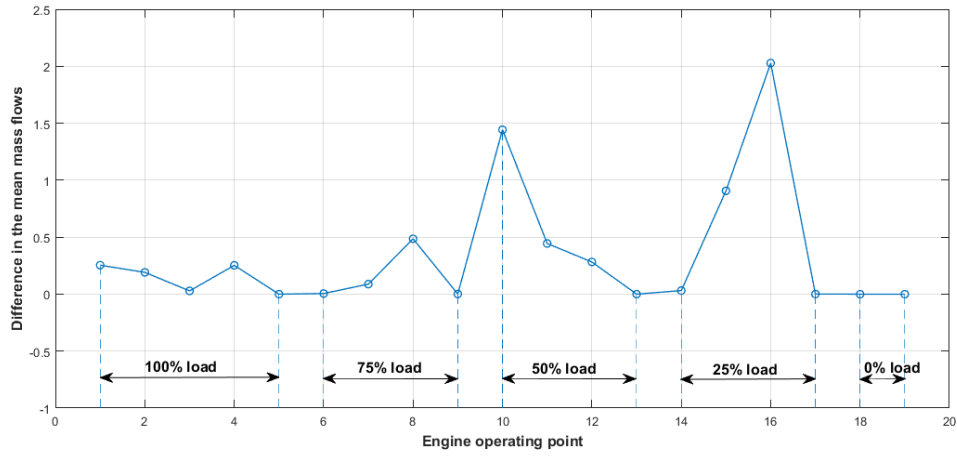


Figure 45: Difference in the mean mass flows of EGR model with case 1 inputs and real time EGR model

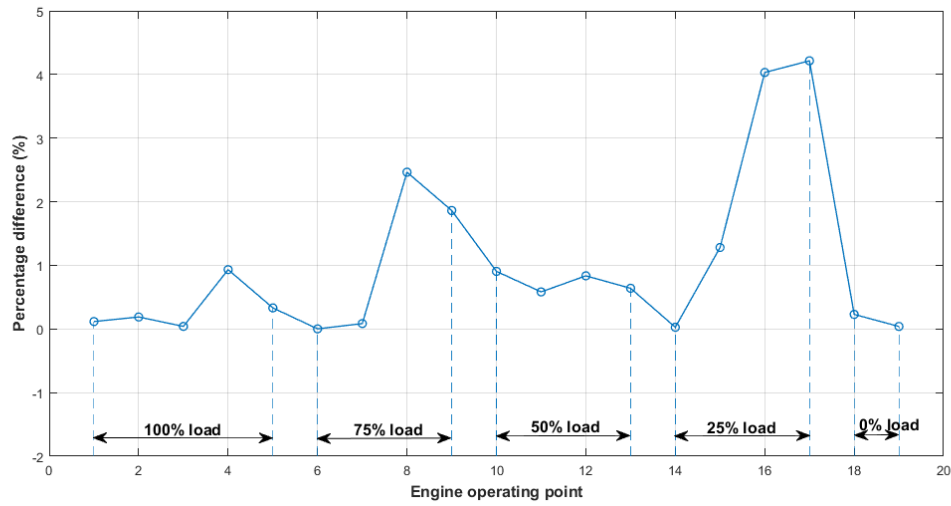


Figure 46: Percentage difference in the mean mass flows of EGR model with case 1 inputs and real time EGR model

7 Real time EGR model- AVL solution

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7.1 Comparison of real time model developed in section 6 and AVL real time solution

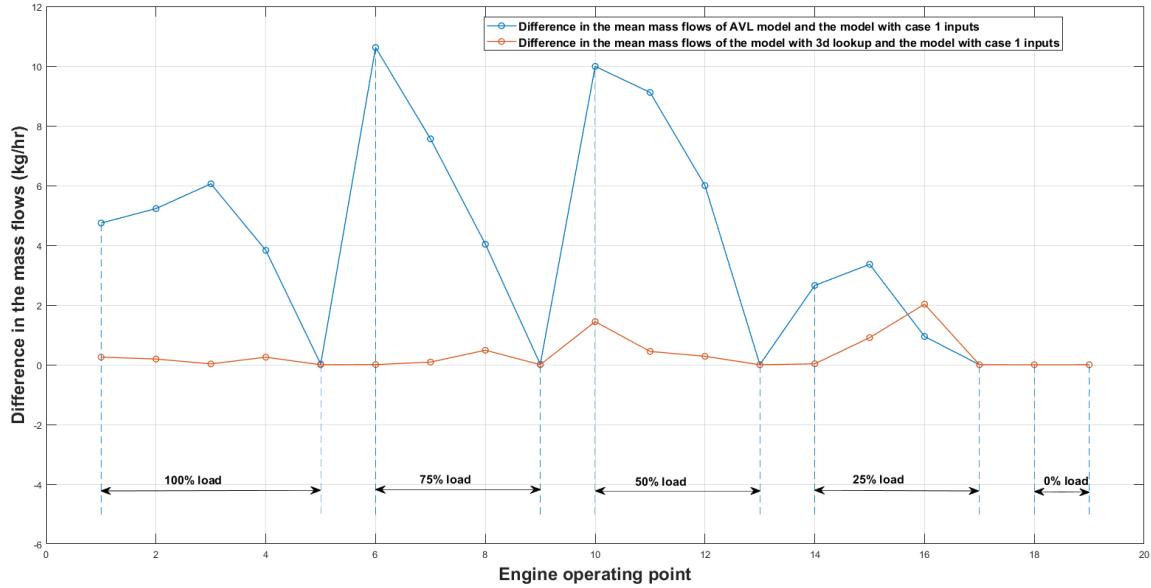


Figure 47: Comparison of the accuracy of the two real time capable models

The above figure 47 shows the difference between the mean mass flow calculated by the real time solution with 3D lookup table and EGR model with case 1 inputs, shown by the red line. The figure also highlights the difference between the mean mass flow calculated by the AVL real time solution and the EGR model with case 1 inputs, shown by the blue line. It can be deduced from the above figure that the difference for the real time solution with 3D lookup table is less as compared to the difference for AVL real time solution. Hence it can be said that the EGR model with 3D lookup has a greater accuracy than the AVL solution.

8 Conclusion and future work

In this thesis, various factors influencing the mass flow through EGR valve of an engine have been analysed. It has been deduced that the upstream pressure of EGR valve is the dominant factor in influencing the mass flow through EGR as compared to other factors like downstream pressure and upstream temperature. It is possible to achieve the reference mean mass flow by an EGR model in which the upstream pressure is modelled as a sinusoidal input, while all other inputs are modelled to be constant. A real time EGR model is developed by modifying an EGR model with sinusoidal upstream pressure as an input. The comparison between the newly developed real time model with 3d lookup table and AVL real time model highlights that the newly developed model has a better accuracy.

One of the important findings of this thesis is that, the amplitude used for modelling of sinusoidal upstream pressure determines if the mass flow calculated by the model would match the reference mean mass flow. In this work, a trial and error method is used to discover the suitable amplitudes for sinusoidal upstream pressures. Future work in this field would be to discover a method to model the amplitude of P_{us} .

Nomenclature

Notations

P_{us}	Pressure on the upstream side of EGR valve	N/m^2
P_{ds}	Pressure on the downstream side of EGR valve	N/m^2
T_{us}	Temperature on the upstream side of EGR valve	K
R	Gas constant	$J/kg.K$
A_{geom}	Flow area	m^2
k	Ratio of specific heats (kappa)	-
C_d	Flow Coefficient	-
π	Pressure ratio	-
ψ	Outflow function (Psi)	-
π_{tr}	Threshold pressure ratio	-
f	Frequency or number of oscillations per second	Hz
P_{mean}	Mean value of upstream pressure	N/m^2
P_{max}	Maximum value of upstream pressure	N/m^2

Abbreviations

EGR	Exhaust Gas Re-circulation
PM	Particulate Matter
CA	Crank Angle
NO_x	Nitrogen Oxides
AF	Air-Fuel ratio

Bibliography

- [1] Ahmad F. "*Modeling, simulation and control of the air-path of an internal combustion engine*," Université de Technologie de Belfort-Montbéliard, 2013. English. (NNT : 2013BELF0203) (tel-01002113)
- [2] Jääskeläinen, H., Khair, M. "*EGR systems and components*," https://www.dieselnets.com/tech/engine_egr_sys.php, accessed Nov. 2018.
- [3] Hussain, J., Palaniradja, K., Alagumurthi, N., Manimaran, R., "*Effect of Exhaust Gas Recirculation (EGR) on Performance and Emission characteristics of a Three Cylinder Direct Injection Compression Ignition Engine*," Elsevier 2012, <https://doi.org/10.1016/j.aej.2012.09.004>.
- [4] Katrašnik, T., "*Transient Momentum Balance—A Method for Improving the Performance of Mean-Value Engine Plant Models*," *Energies* 2013, 6, 2892-2926; doi:10.3390/en6062892
- [5] CRUISE™ M User Manual (Version R2017f), Component Libraries/Gas Flow/Restriction/Theory/Transient Momentum Balance, AVL, accessed Aug. 2018.
- [6] Bertrand, D.V., AVL Software and Functions GmbH, personal communication, Sep. 2018.

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